IPQ8064 Network Processor

Device Specification

80-Y5080-3 Rev. E

February 28, 2014

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Revision history

Revision	Date	Description
A	September 2013	Initial release
B	December 2013	1. Remove MDIO and MDC signals. 2. Clarify the QFPROM footnote in the Power-On Sequence table. 3. PCIE1_CLK_REQ should be PCIE1_CLK_REQ_N 4. Add order number. 5. Add Electrical Specifications. 6. Remove Boot_Config option 00. 7. Specified the supported audio formats
C	February 2014	1. Correct NSS frequency in block diagram to 733 MHz.
D	February 2014	1. Specify all calibration and reference resistances 2. Add timing diagrams and data for peripheral interfaces 3. Add Section 3.4 “Digital logic characteristics” 4. Add Chapter 5 “Carrier, Storage, and Handling” 5. Add Chapter 6 “PCB Mounting Guidelines” 6. Add Chapter 7 “Part Reliability” 7. QSGMII_SEL_25M is 1.1 V 8. Add L1 cache sizes 9. Update PVS values in Table 3-3
E	February 2014	1. Minor format change; no content change.

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1 Introduction

1.1 IPQ8064 introduction

The IPQ8064 is an internet processor for carrier gateway and smart retail router. The IPQ8064 includes a powerful dual-core SMP Krait CPU at 1.4 GHz with an ARM-v7-compliant instruction set. Each Krait CPU includes a 128-bit SIMD DSP, referred to as VeNum. The CPU system is supported by 1 MByte of L2 cache as well as 16 KiB L1 I-Cache and 16 KiB L1 D-Cache.

The IPQ8064 also includes a network accelerator with an aggregate throughput capacity of 5 Gbps. This network accelerator is referred to as the Networking Subsystem (NSS). The NSS consists of two programmable, multithreaded, network accelerator engines. Each network accelerator engine supports 12 threads of execution. The function of each thread is determined by firmware (FW) provided by Qualcomm Atheros. The NSS architecture allows the threads to operate in parallel. NSS functions include L2/L3 routing/bridging, Samba acceleration, LRO/TSO offload, connectivity acceleration, as well as traffic management.

The IPQ8064 typically runs an OpenWRT Linux IP stack on the Dual Core SMP Krait, while the NSS accelerates networking traffic (fast path). The NSS operates under full control of the Linux stack. Rules in the NSS are set by the Linux stack. Networking packets that do not match these rules are passed on to the Linux stack through an exception handling mechanism in the NSS.

To support a large variety of platforms, the IPQ8064 is equipped with several high speed Serdes and general purpose interfaces. These include:

- 4×GMACs, configurable as:
  - 1×QSGMII
  - 3×SGMII and 1×RGMII
  - 2×SGMII and 2×RGMII
- 3×PCIe 2.0 (root complex)
- 2×USB3 (host and device mode)
- SATA3
- 32-bit DDR3/1066
- 8-bit NAND/SDIO
  - ECC 8b/512, 16b/1024
- 4-bit SDIO
- HSIC
- 2×Transport Stream Interface (TSIF)

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- 6×general-purpose serial interfaces (GSBI), configurable as SPI, UART, I2C, UIM.
- JTAG

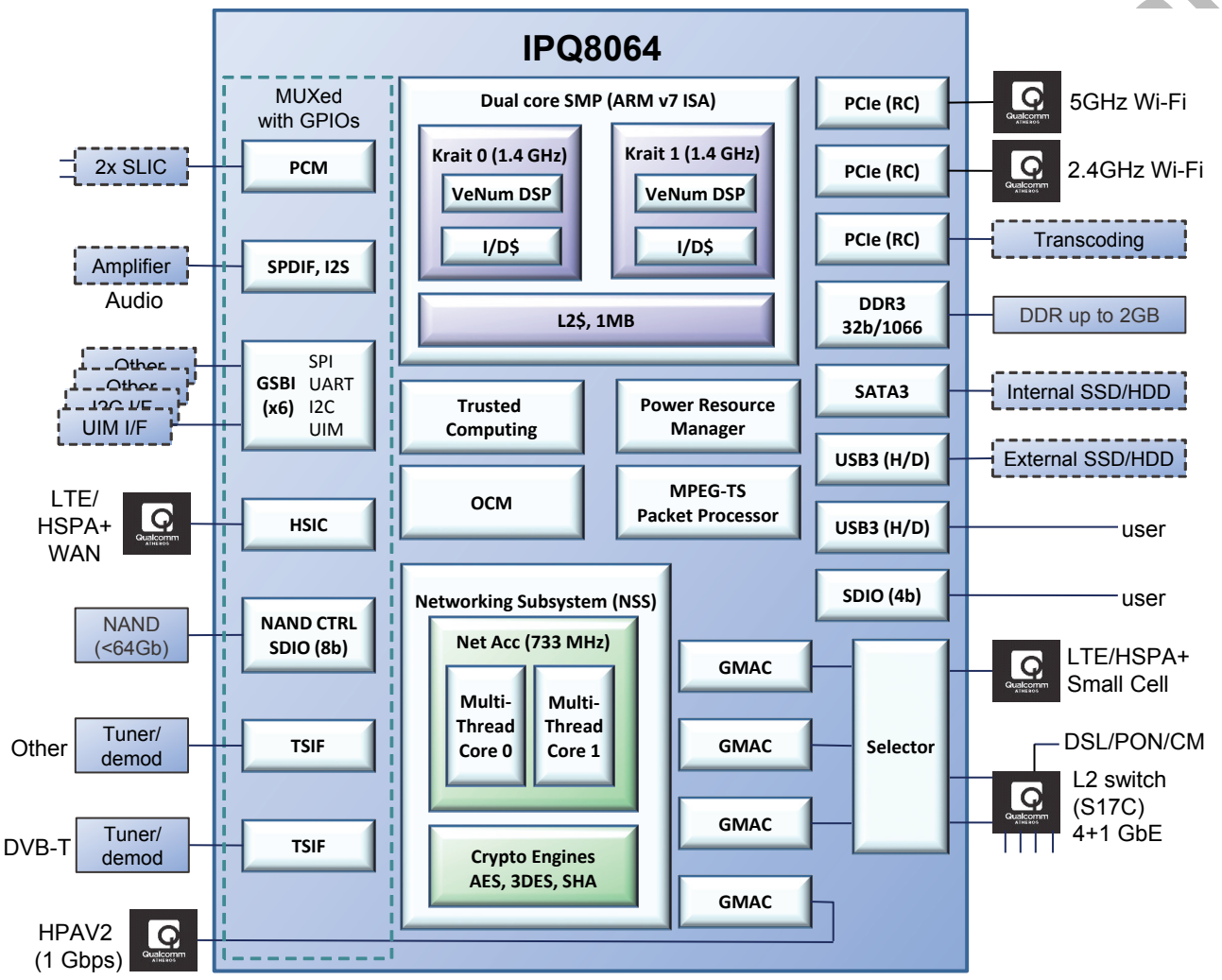


Figure 1-1 IPQ8064 functional block diagram

Complementary blocks within the IPQ8064 include:

- TrustZone, supplemented with additional trusted-computing logic at each bus interface
- Resource power manager
- MPEG-TS Packet Processor
- On-chip memory (OCM)
- NSS network accelerator
- Security acceleration for AES, 3DES, SHA

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1.2 IPQ8064 features

1.2.1 Architecture and processing units

1.2.1.1 Kraits

The primary application processors are dual-core SMP Kraits running at up to 1.4 GHz. The applications processors are ARMv7 compliant with Trustzone support. They include a vector floating-point accelerator, implementing an extended version of the Neon Instruction Set. The applications processors are supported by a cache hierarchy, consisting of:

- L2: 1 MB
- L1: 16 KB I-cache, 16 KB D-cache
- L0: 4 KB I-cache, 4 KB D-cache

1.2.1.2 Network Subsystem (NSS)

The Network Subsytem (NSS) is an advanced engine for accelerating the processing of network packets. It consists of two multithreaded, network-accelerator cores up to 733 MHz, with 16 KB L1 instruction cache, 32 KB L1 data cache, 192 KB on-chip memory (OCM).

Each network-accelerator core is a single-issue, in-order, 12-way multithreaded core. Each thread maintains a fully independent execution context. The network-accelerator cores share 128 KB memory for interprocessor communication and for storing I/O descriptors, and cryptography descriptors and commands.

1.2.1.3 Crypto Engines

The IPQ8064 contains cryptography engines, providing high-speed encryption, decryption, and authentication. Computing a signature for authentication can be performed simultaneously with encryption or decryption.

The crypto engines support the following algorithms:

- Encryption algorithms
  - AES (128 and 256 bit key support) (XTS, CTR, CCM, CMAC, CBC and ECB modes)
  - DES/3DES (CBC and EBC modes)
- Authentication algorithms
  - SHA1
  - SHA224 (the result of supporting SHA256)
  - SHA256
  - HMAC-SHA1
  - HMAC-SHA2

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The DMA engine in the crypto engine is configured to support four pipe pairs. Pipe pairs allow the crypto engine to support multiple IPsec streams.

Crypto offload is available for various applications (DRM, for example). Also, encryption and decryption of content can be offloaded to the NSS for supported algorithms (IPsec, for example).

1.2.1.4 Security controller and QFPROM

The Qualcomm Fuse Programmable Read Only Memory (QFPROM) is a large fuse array.

- Handles chip-wide configuration for security and debugging; provides persistent storage of ID numbers and sensitive key data.
- Secure debug facility that creates a secure tester channel for provisioning key information or enabling features and debug capabilities; also provides a gateway for all software and JTAG accesses to the QFPROM.
- Primary and secondary hardware key blocking for the secure file system

1.2.2 Ethernet

1.2.2.1 GMAC

The NSS includes four 10/100/1000 gigabit MAC Controllers that are IEEE 802.3-2008 compliant. Each controller features:

- IP checksum offload with 9600 jumbo frames.
- IEEE 1588-2 Time synchronization.
- DMA engine

The four controllers can operate in any of the configurations shown in Table 1-1.

Table 1-1 GMAC configurations

Configuration	GMAC 1	GMAC 2	GMAC 3	GMAC 4
A	QSGMII			
B	RGMII (2.5 V)	SGMII	SGMII	SGMII
C	RGMII (2.5 V)	RGMII (1.8 V)	SGMII	SGMII

In configuration A, all four GMAC controllers are operating across a single QSGMII link. In configuration B, up to three GMAC controllers can be operating on an SGMII link each, while a single GMAC controller is on a 2.5 V RGMII link. In configuration C, up to two GMAC controller can be operating on an RGMII link each, and each of up to two GMAC controllers can be operating on an SGMII link.

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1.2.2.2 QSGMII/SGMII

The QSGMII/SGMII PHY is an analog core that supports QSGMII/SGMII external interface. It can support either a single QSGMII interface or up to three independent SGMII interfaces. The core is composed of a combo QSGMII/SGMII SerDes and two SGMII SerDes.

- QSGMII is a 5 Gbps SerDes that is compliant with Cisco Systems QSGMII Specification Revision 1.2.
- SGMII a 1.25 Gbps SerDes that is compliant with Cisco Systems SGMII Specification Revision 1.8.

The QSGMII/SGMII core also sources a reference clock for the PCIe, SATA 3.0, and USB 3.0/2.0 units, as well as the Ethernet subsystem.

1.2.3 Dedicated peripheral interfaces

1.2.3.1 PCIe 2.0

The IPQ8064 offers three single-lane PCIe 2.0 ports in the root complex (RC) mode. These can be used for external Wi-Fi chipset, external transcryption, xDSL, etc.

Each PCIe offers 500 MB/s data rate using a 5 GHz SERDES interface. The protocol allows for backward compatibility between PCIe 1.0 and 2.0.

1.2.3.2 USB

Two USB3.0 ports are available to connect to SuperSpeed USB3 devices, such as SSD or HDD storage, as well as to legacy USB2 devices. Each port provides a single root-hub for USB2 and for USB3. The cores support both host and device mode.

1.2.3.3 SATA

One SATA3 host port is available for local HDD. The port is backward compatible to SATA1 and SATA2. The IPQ8064 can support eSATA with an external eSATA chip.

1.2.3.4 SDC3 4-bit Secure Digital Card controller

The Secure Digital Card Controller (SDCC4) supports up to SD3.0 specification devices as well as SDIO and MMC, and is fully backward compatible.

- 4-bit for external SD/MMC card, for connection to external SD-card devices and SDIO devices.

1.2.4 Programmable peripheral interfaces

The pins for several I/O ports are muxed with GPIOs under control of the Top-Level Mode Multiplexer (TLMM) block. Configuration of the TLMM is managed by software. Availability of

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a given peripheral depends on configuration. Pins not used for specific peripheral blocks are available as general-purpose I/Os.

1.2.4.1 PCM

The PCM interface enables two-way communication of raw audio to an external SLIC chip. The Krait processors can run a SIP VoIP stack that drives the PCM interface. The external SLIC ICs are used to create the ring, on-hook and off-hook signal; as well as, provide DAC/ADC functionality for the voice signal. The output of the SLIC chip is the two wire interface that connects to a regular analog POTS phone. One of the SPI interfaces can provide the control channel for the SLIC IC.

1.2.4.2 SPDIF/I^{2}S Audio

This audio block contains the DMA engines and interface for SPDIF, as well as the I^{2}S interfaces. The I^{2}S interfaces support either four stereo output channels or three stereo output channels and one stereo input channel. SPDIF input is not supported. The supported audio formats are:

- SPDIFTX 24-bit RAW
- SPDIFTX 16-bit compressed AC-3 format in 2.0 and 5.1 channels
- MI2S 16-bit and 32-bit for 5.1 and 7.1 channels
- MI2S 24-bit (stereo) for 5.1 and 7.1 channels

The Krait processors can be used in some applications to encode or decode audio data. For example:

- Krait based application downloads MP3 encoded data from the Internet and decodes it. The raw audio samples generated by the decoder are then sent to an external DAC and audio amplifier over the I^{2}S interface.
- Krait accepts raw audio data from an ADC connected to an analog audio source such as a music player. The raw audio data received by Krait is then transmitted over the LAN to various speakers in the home.

1.2.4.3 HSIC

The High-Speed Inter-Chip (HSIC) supplement to the USB 2.0 specification provides a better solution for USB chip-to-chip interconnect by removing the analog transceivers, thus reducing complexity. The HSIC uses 240 MHz DDR (Double Data Rate) signaling to provide high speed of 480Mbps. The HSIC in IPQ8064 is host only. High-Speed 480 Mbps USB transfers are 100% host driver compatible with traditional USB. All features of High-Speed Inter-Chip USB specification are supported.

1.2.4.4 GSBI

Each of the six General-Purpose Serial Bus Interface units can be configured by software to support the serial protocols UART, UIM, I2C, and SPI. At any given time, one of the serial protocols may be selected for use in each unit; in addition, there are two concurrency schemes that

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allow for a two-wire UART and I2C or UIM and I2C to be used simultaneously. Refer to Table 2-34 for configuration details.

1.2.4.5 SDC1 8-bit Secure Digital Card controller

The Secure Digital Card Controller supports up to SD3.0 specification devices as well as SDIO and MMC, and is fully backward compatible.

- 8-bit for external eMMC storage or boot memory.

1.2.4.6 NAND Flash (EBI2) Interface

The EBI2 interface supports NAND flash for boot memory or storage.

- 1.8 V memory devices
- Multiplexed address/data bus
- Runs at up to 64 MHz
- 8-bit NAND flash

1.3 Terms and acronyms

Table 1-2 defines terms and acronyms commonly used throughout this document.

Table 1-2 Terms and acronyms

Term	Definition
CM	Cable Modem
DDR	Double data rate
DECT	Digital Enhanced Cordless Telecommunications
DRM	Digital Rights Management
DSL	Digital Subscriber Line
DSP	Digital Signal Processor
DVB-T	Digital Video Broadcasting – Terrestrial
eMMC	Embedded MMC
ETM	Embedded trace macrocell
GMII	Gigabit MII
GPIO	General-purpose input/output
GSBI	General Serial Bus Interface
HDD	Hard Disk Drive
HPAV2	HomePlug AV2 ()
HSIC	High-Speed Inter-Chip (USB chip-to-chip interconnect)
HSPA+	Evolved High-Speed Packet Access

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Table 1-2 Terms and acronyms (cont.)

Term	Definition
HSUART	High-Speed UART
I/D$	Instruction/Data Cache
I^2C	Inter-integrated circuit
I^2S	Inter-IC sound
IRDA	Infrared Data Association
JTAG	Joint Test Action Group
LNA	Low-Noise Amplifier
LTE	Long-Term Evolution
MI2S	Multichannel I2S
MII	Media-Independent Interface
MMC	MultiMediaCard
MPEG	Moving Picture Experts Group
MSL	Moisture Sensitivity Level
NAND	NAND flash memory
NOR	NOR flash memory
NSS	Network Subsystem
OCM	On-chip Memory
PA	Power Amplifier
PCIe	Peripheral Component Interconnect Express
PCM	Pulse-Code Modulation
PMIC	Power management integrated circuit
PON	Passive Optical Network
POTS	Plain old telephone service
PPD	Peak-to-peak differential
PRNG	Pseudo-Random Number Generator
QSGMII	Quad SGMII
RC	Root Complex
RF	Radio Frequency
RGMII	Reduced GMII
RPM	Resource and Power Manager
RUIM	Removable UIM
SATA	Serial ATA (AT Attachment)
SDIO	Secure Digital Input Output
SDMMC (SD/MMC)	Secure Digital MultiMediaCard
SGMII	Serial GMII
SLIC	Subscriber line interface card

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Table 1-2 Terms and acronyms (cont.)

Term	Definition
SMP	Symmetric Multiprocessing
SPDIF (S/PDIF)	Sony/Philips Digital Interconnect Format
SPI	Serial Peripheral Interface (Bus)
SPMI	System power management interface
SSBI	Single-wire serial bus interface
SSD	Solid-State Drive
TAP	Test Access Port
TSIF	Transport Stream Interface
UART	Universal Asynchronous Receiver/Transmitter
UI	Unit interval
UIM	User identity module
USB	Universal Seral Bus
VOIP	Voice over IP
WAN	Wide Area Network

1.4 Special marks

Table 1-3 defines special marks used in this document.

Table 1-3 Special marks

Mark	Definition
[ ]	Brackets ([ ]) sometimes follow a pin, register, or bit name. These brackets enclose a range of numbers. For example, EBI2_AD_[7:4] may indicate a range that is 4 bits in length, or DATA[7:0] may refer to all eight DATA pins.
_N	A suffix of _N indicates an active low signal. For example, RESIN_N.
0x0000	Hexadecimal numbers are identified with an x in the number, for example, 0x0000. All numbers are decimal (base 10), unless otherwise specified. Non-obvious binary numbers have the term binary enclosed in parentheses at the end of the number, for example, 0011 (binary).
|	A vertical bar in the outside margin of a page indicates that a change was made since the previous revision of this document.

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2 Pin Definitions

2.1 I/O parameter definitions

Table 2-1 I/O description (pad type) parameters

Symbol	Description
Pad attribute	
A	Analog signal (does not contain pad circuitry)
AI	Analog input (does not include pad circuitry)
AO	Analog output (does not include pad circuitry)
AIO	Analog input/output (does not contain pad circuitry)
B	Bidirectional digital with CMOS input
BS	CMOS bidirectional I/O with Schmitt trigger input
HVIN	Pad uses the pad voltage input to the core, core_in_px_*
I	CMOS input; pad's output enable tied off disabled in functional mode
IE	Pad uses the input enable signal, core_ie_*
IS	CMOS input with Schmitt trigger
O	CMOS output; pad's input enable tied off disabled in functional mode
W	Wakeup interrupt during deep-sleep mode
Z	CMOS tri-state output; pad's input enable tied off disabled in functional mode
V_P	Pad power
G_P	Pad ground
V_C	Core power
G_C	Core ground
V_A	Analog power
G_A	Analog ground
Pad-pull details for digital I/Os	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters and is a prefix to other programmable options: NP: pdpukp = default no-pull with programmable options following the colon (:) PD: nppukp = default pull-down with programmable options following the colon (:) PU: nppdkp = default pull-up with programmable options following the colon (:) KP: nppdpu = default keeper with programmable options following the colon (:)
KP	Contains an internal weak keeper device (keepers cannot drive external buses)

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Table 2-1 I/O description (pad type) parameters (cont.)

Symbol	Description
NP	Contains no internal pull
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device
Pad voltage groupings	
P1	Pad group 1 (EBI1); tied to VDDPX_1 pins (1.35 V only)
P2	Pad group 2 (SDC3); tied to VDDPX_2 pins (1.8 V or 2.95 V)
P3	Pad group 3 (most peripherals); tied to VDDPX_3 pins (1.8 V only)
P4	Pad group 4 (HSIC); tied to VDDPX_4 pins (1.2 V or 1.8 V)
P5	Pad group 5 (UIM); tied to VDDPX_5 pins (1.8 V or 2.95 V)
P6	Pad group 6 (RGMII); tied to VDDPX_6 pins (2.5 V)
Output current drive strength	
EBI1 pads	Pads for EBI1 are tailored for 1.2 V interfaces and are source terminated. At the pads, after the source termination, the drive strength at I_{OL}, I_{OH} is equivalent to a range of 0.8 to 2.5 mA in nonlinear steps when the JEDEC standard range (90%/10%) is followed.
3.0 V (H) pads	Programmable drive strength, 2 to 8 mA in 2 mA steps
Others [1]	Programmable drive strength, 2 to 16 mA in 2 mA steps

1. Digital pads other than EBI1 pads, or high-voltage tolerant pads.

2.2 IPQ8064 Pin map

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Table 2-2 IPQ8068 BGA519 Pin assignments (top view)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A		VSSX_0	EBI1_DQ24	EBI1_DQ25	RGMII_RX_0	RGMII_CK_RX	RGMII_TX_0	RGMII_TX_2	GPIO_20	GPIO_17	GPIO_66	PCIE3_HSI_0_N	PCIE3_HSO_0_N	PCIE3_REFCLK_P	PCIE2_HSI_0_N	PCIE2_HSO_0_N	PCIE2_REFCLK_P	PCIE1_HSI_0_N	PCIE1_HSO_0_N	PCIE1_REFCLK_P	PCIE2_CLK_OUT_N	PCIE3_CLK_OUT_N	PCIE1_CLK_OUT_N	GPIO_31	GPIO_29	GPIO_55	VSSX_0		A
B	VSSX_0	VSSX_0	EBI1_DQ29	EBI1_DQ26	RGMII_CTL_RX	RGMII_RX_2	RGMII_CTL_TX	RGMII_TX_3	GPIO_02	GPIO_16	GPIO_14	PCIE3_HSI_0_P	PCIE3_HSO_0_P	PCIE3_REFCLK_N	PCIE2_HSI_0_P	PCIE2_HSO_0_P	PCIE2_REFCLK_N	PCIE1_HSI_0_P	PCIE1_HSO_0_P	PCIE1_REFCLK_N	PCIE2_CLK_OUT_P	PCIE3_CLK_OUT_P	PCIE1_CLK_OUT_P	GPIO_30	GPIO_28	GPIO_56	VSSX_0	VSSX_0	B
C	EBI1_DM2	EBI1_DQ31		VDDPX_6	VDDPX_6	RGMII_RX_1	RGMII_RX_3	RGMII_TX_1	RGMII_CK_TX	GPIO_19	GPIO_15	VSSX_0	VSSA_PCIE3	VDDA_VP_PCIE3	VDDA_VPTX0_PCIE3	VSSA_PCIE2	VDDA_VPTX0_PCIE2	VDDA_VPTX0_PCIE1	VSSA_PCIE1	GPIO_62	GPIO_61	GPIO_59	GPIO_27	GPIO_32	VDDPX_3	GPIO_57	VSSX_0	QSGMII_XTLI	C
D	EBI1_DQSB3	EBI1_DQS3	VSSX_0		EBI1_DQ30	EBI1_DQ27	VDDPX_VBIAS_RGMII	VSSX_0	VDDPX_3	GPIO_21	GPIO_18	VSSX_0	VSSA_PCIE3	PCIE3_REXT	VSSA_PCIE2	PCIE2_REXT	VDDA_VP_PCIE2	VDDA_VP_PCIE1	PCIE1_REXT	VSSA_PCIE1	GPIO_60	GPIO_52	GPIO_51	GPIO_33		GPIO_58	VDDA_QSGMII_PLL	QSGMII_XTLO	D
E	EBI1_DQSB2	EBI1_DQS2	VDDPX_1	EBI1_DQ28																					VDDPX_3	VDDPX_3	VDDA_QSGMII_DRV	VSSA_QSGMII	E
F	EBI1_DQ19	EBI1_DQ18	EBI1_DQ16	EBI1_DM3																					VSSX_0	VSSA_QSGMII	QSGMII_QSO0_N	QSGMII_QSO0_P	F
G	EBI1_DQ23	EBI1_DQ21	EBI1_DQ20	EBI1_DQ17																					QSGMII_SEL_25M	VDDA_QSGMII_TX	QSGMII_QSI0_N	QSGMII_QSI0_P	G
H	EBI1_A09	EBI1_A08	EBI1_DQ22	VDDPX_1																					QSGMII_RESREF	VDDA_QSGMII_RX	SGMII_SO1_N	SGMII_SO1_P	H
J	EBI1_A05	EBI1_A06	EBI1_A07	VSSX_0					VDDCX_1	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VDDCX_1	VSSX_0	VDDA_VPH_PCIE3	VDDA_VPH_PCIE2	VDDA_VPH_PCIE1	VDDCX_1	VSSA_QSGMII					VSSA_QSGMII	VDDA_QSGMII_PLL	SGMII_SI1_N	SGMII_SI1_P	J
K	EBI1_A12	EBI1_A13	EBI1_A11	VDDA_CDC_EBI1_1					VDDCX_1	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VDDCX_1	VSSX_0	VSSX_0	VSSA_PCIE2	VDDCX_1	VDDCX_1	VSSA_QSGMII					QSGMII_CLK25M_OUT	VDDA_QSGMII_H18	SGMII_SI2_N	SGMII_SI2_P	K
L	EBI1_A02	EBI1_A03	EBI1_A10	VREF_EBI1_1					VSSX_0	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_QSGMII	VDDA_SGMII					VSSA_QSGMII	VDDA_SGMII	SGMII_SO2_N	SGMII_SO2_P	L
M	EBI1_A00	EBI1_A14	EBI1_A01	EBI1_ODT1					VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_QSGMII	VSSA_QSGMII					RSVD	RSVD	VDDCX_1	VSSX_0	M
N	EBI1_BA1	EBI1_BA0	EBI1_BA2	VDDPX_1					VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	GPIO_00					VSSX_0	VDDPX_3	RSVD	RSVD	N
P	EBI1_CAL	EBI1_A04	VSSX_0	VSSX_0					VDDPX_3	VDDCX_1	VSSX_0	VSSX_0	VSS_HF_PLL12	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	GPIO_01					RSVD	VDDCX_1	RSVD	RSVD	P
R	EBI1_CK0B	EBI1_CK0	VSSX_0	VSSX_0					VDDA_CDC_EBI1_CA	VSSX_0	VDDCX_1	VDD_HF_PLL12	VDD_HF_PLL12	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_USB3_SS	VDDD_USB3_SS					VDDD_USB3_SS_VPTX	USB3_SS_RESREF	VSSA_USB3_SS	VSSA_USB3_SS	R
T	EBI1_WEN	EBI1_CKE0	EBI1_CS1N	VDDPX_1					VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_USB3_HS	VDDD_USB3_HS					VDDA_USB3_SS_1P8	USB3_HS_VBUS	USB3_SS_TX0M	USB3_SS_TX0P	T
U	EBI1_CS0N	EBI1_RESETN	EBI1_ODT0	EBI1_CKE1					VSSX_0	VSSX_0	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	VDD_APC1	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0					VDDA_USB3_HS_1P8	VDDA_USB3_HS_3P3	USB3_SS_RX0M	USB3_SS_RX0P	U
V	EBI1_CASN	EBI1_RASN	EBI1_DQ08	VREF_EBI1_0					VDDCX_1	VSSX_0	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	VDD_APC1	VSSX_0	VDDCX_1	VDDCX_1	VSSA_USB1_SS	VDDD_USB1_SS					VSSA_USB3_HS	USB3_HS_TXRTUNE	USB3_HS_DM	USB3_HS_DP	V
W	EBI1_DQ09	EBI1_DQ10	EBI1_DQ11	VDDA_CDC_EBI1_0					VDD_HF_PLL9	VSS_HF_PLL9	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	VDD_APC1	VSSX_0	VDD_QFPROM_BLOW	VDDCX_1	VSSA_USB1_HS	VDDD_USB1_HS					USB3_HS_SYSCLOCK	USB3_HS_ID	USB1_3_SS_REF_CLKP	USB1_3_SS_REF_CLKM	W
Y	EBI1_DQ12	EBI1_DQ13	EBI1_DQ15	VSSX_0					VDDCX_1	VDDCX_1	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	VDD_APC1	VSS_HF_PLL10	VSS_SR2_PLL	VSS_SR_PLL0	VSSX_0	VSSX_0					VDDD_USB1_SS_VPTX	VSSA_USB1_SS	USB1_SS_TX0M	USB1_SS_TX0P	Y
AA	EBI1_DQSB1	EBI1_DQS1	EBI1_DM1	VDDPX_1																					USB1_SS_RESREF	VDDA_USB1_SS_1P8	USB1_SS_RX0M	USB1_SS_RX0P	AA
AB	EBI1_DQ14	EBI1_DM0	EBI1_DQ00	EBI1_DQ01																					VSSA_USB1_HS	VDDA_USB1_HS_1P8	USB1_HS_VBUS	USB1_HS_SYSCLOCK	AB
AC	EBI1_DQS0	EBI1_DQSB0	EBI1_DQ03	EBI1_DQ05																					USB1_HS_TXRTUNE	USB1_HS_ID	USB1_HS_DM	USB1_HS_DP	AC
AD	EBI1_DQ02	EBI1_DQ04	VDDPX_1	EBI1_DQ07																					VDDA_USB1_HS_3P3	VDDPX_3	VSSX_0	VSSX_0	AD
AE	EBI1_DQ06	TRST_N	VDDPX_1	VSS_SENSE_APC_0_1	VDD_SENSE_APC_0	VDD_SENSE_APC_1	VDDPX_3	VSSX_0	GPIO_43	GPIO_47	VDDPX_3	VSSX_0	VDD_CDC_SDC3	VSSX_0	VDDPX_3	VSSX_0	VDD_HF_PLL10	VDD_SR2_PLL	VDD_SR_PLL	VDDPX_5	VSSX_0	VDDPX_3	GPIO_10			VDDPX_3	GPIO_48	GPIO_49	AE
AF	RTCK	TMS	VSSX_0	VDDPX_3	VDDPX_3	TDO	MODE_0	GPIO_37	GPIO_34	GPIO_44	GPIO_46	SDC3_DATA_1	VDDPX_VBIAS_SDC	VDD_CDC_SDC1	VSSX_0	VDDPX_3	GPIO_12	NC	NC	SLEEP_CLK	GPIO_65	GPIO_63	VDDPX_VBIAS_UIM	VDDPX_4	VDDPX_4	VSSX_0	GPIO_50	GPIO_54	AF
AG	VSSX_0	VSSX_0	TDI	TCK	MODE_1	GPIO_26	GPIO_40	GPIO_36	GPIO_42	GPIO_45	SDC3_CMD	SDC3_CLK	SDC3_DATA_3	VDDPX_2	VSSX_0	GPIO_04	GPIO_13	NC	SLEEP_CLK_32K_OUT	NC	CXO	GPIO_64	GPIO_68	GPIO_06	GPIO_08	GPIO_53	VSSX_0	VSSX_0	AG
AH		VSSX_0	RESIN_N	RESOUT_N	SRST_N	GPIO_35	GPIO_39	GPIO_41	GPIO_38	SDC3_DATA_0	SDC3_DATA_2	GPIO_24	GPIO_24	GPIO_23	GPIO_22	GPIO_05	GPIO_03	GPIO_11	CLK_25M_OUT	NC	NC	NC	GPIO_67	USB2_HSIC_CAL	GPIO_09	GPIO_07	VSSX_0		AH
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	

EBI1* PCI* USB* GMII* GPIO* VDDPX* VDDCX* VDD_APC* *VDD* *VSS* VSSX_0

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Table 2-3 IPQ8068 BGA519 Pin assignments (upper left quadrant)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A		VSSX_0	EBI1_DQ24	EBI1_DQ25	RGMII_RX_0	RGMII_CK_RX	RGMII_TX_0	RGMII_TX_2	GPIO_20	GPIO_17	GPIO_66	PCIE3_HSI_0_N	PCIE3_HSO_0_N	PCIE3_REFCLK_P	A
B	VSSX_0	VSSX_0	EBI1_DQ29	EBI1_DQ26	RGMII_CTL_RX	RGMII_RX_2	RGMII_CTL_TX	RGMII_TX_3	GPIO_02	GPIO_16	GPIO_14	PCIE3_HSI_0_P	PCIE3_HSO_0_P	PCIE3_REFCLK_N	B
C	EBI1_DM2	EBI1_DQ31		VDDPX_6	VDDPX_6	RGMII_RX_1	RGMII_RX_3	RGMII_TX_1	RGMII_CK_TX	GPIO_19	GPIO_15	VSSX_0	VSSA_PCIE3	VDDA_VP_PCIE3	C
D	EBI1_DQSB3	EBI1_DQS3	VSSX_0		EBI1_DQ30	EBI1_DQ27	VDDPX_VBIAS_RGMII	VSSX_0	VDDPX_3	GPIO_21	GPIO_18	VSSX_0	VSSA_PCIE3	PCIE3_REXT	D
E	EBI1_DQSB2	EBI1_DQS2	VDDPX_1	EBI1_DQ28											E
F	EBI1_DQ19	EBI1_DQ18	EBI1_DQ16	EBI1_DM3											F
G	EBI1_DQ23	EBI1_DQ21	EBI1_DQ20	EBI1_DQ17											G
H	EBI1_A09	EBI1_A08	EBI1_DQ22	VDDPX_1											H
J	EBI1_A05	EBI1_A06	EBI1_A07	VSSX_0					VDDCX_1	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VDDCX_1	J
K	EBI1_A12	EBI1_A13	EBI1_A11	VDDA_CDC_EBI1_1					VDDCX_1	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VDDCX_1	K
L	EBI1_A02	EBI1_A03	EBI1_A10	VREF_EBI1_1					VSSX_0	VSSX_0	VDD_UBI32	VDD_UBI32	VSSX_0	VSSX_0	L
M	EBI1_A00	EBI1_A14	EBI1_A01	EBI1_ODT1					VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	M
N	EBI1_BA1	EBI1_BA0	EBI1_BA2	VDDPX_1					VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	N
P	EBI1_CAL	EBI1_A04	VSSX_0	VSSX_0					VDDPX_3	VDDCX_1	VSSX_0	VSSX_0	VSS_HF_PLL12	VDDCX_1	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

EBI1* PCI* USB* | GMII* | GPIO* | VDDPX* | VDDCX* | VDD_APC* | *VDD* | *VSS* | VSSX_0

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Table 2-4 IPQ8068 BGA519 Pin assignments (upper right quadrant)

	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A	PCIE2_HSI_0_N	PCIE2_HSO_0_N	PCIE2_REFCLK_P	PCIE1_HSI_0_N	PCIE1_HSO_0_N	PCIE1_REFCLK_P	PCIE2_CLK_OUT_N	PCIE3_CLK_OUT_N	PCIE1_CLK_OUT_N	GPIO_31	GPIO_29	GPIO_55	VSSX_0		A
B	PCIE2_HSI_0_P	PCIE2_HSO_0_P	PCIE2_REFCLK_N	PCIE1_HSI_0_P	PCIE1_HSO_0_P	PCIE1_REFCLK_N	PCIE2_CLK_OUT_P	PCIE3_CLK_OUT_P	PCIE1_CLK_OUT_P	GPIO_30	GPIO_28	GPIO_56	VSSX_0	VSSX_0	B
C	VDDA_VPTX0_PCIE3	VSSA_PCIE2	VDDA_VPTX0_PCIE2	VDDA_VPTX0_PCIE1	VSSA_PCIE1	GPIO_62	GPIO_61	GPIO_59	GPIO_27	GPIO_32	VDDPX_3	GPIO_57	VSSX_0	QSGMII_XTLI	C
D	VSSA_PCIE2	PCIE2_REXT	VDDA_VP_PCIE2	VDDA_VP_PCIE1	PCIE1_REXT	VSSA_PCIE1	GPIO_60	GPIO_52	GPIO_51	GPIO_33		GPIO_58	VDDA_QSGMII_PLL	QSGMII_XTLO	D
E											VDDPX_3	VDDPX_3	VDDA_QSGMII_DRV	VSSA_QSGMII	E
F											VSSX_0	VSSA_QSGMII	QSGMII_QSO0_N	QSGMII_QSO0_P	F
G											QSGMII_SEL_25M	VDDA_QSGMII_TX	QSGMII_QSI0_N	QSGMII_QSI0_P	G
H											QSGMII_RESREF	VDDA_QSGMII_RX	SGMII_SO1_N	SGMII_SO1_P	H
J	VSSX_0	VDDA_VPH_PCIE3	VDDA_VPH_PCIE2	VDDA_VPH_PCIE1	VDDCX_1	VSSA_QSGMII					VSSA_QSGMII	VDDA_QSGMII_PLL	SGMII_SI1_N	SGMII_SI1_P	J
K	VSSX_0	VSSX_0	VSSA_PCIE2	VDDCX_1	VDDCX_1	VSSA_QSGMII					QSGMII_CLK25M_OUT	VDDA_QSGMII_H18	SGMII_SI2_N	SGMII_SI2_P	K
L	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_QSGMII	VDDA_SGMII					VSSA_QSGMII	VDDA_SGMII	SGMII_SO2_N	SGMII_SO2_P	L
M	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_QSGMII	VSSA_QSGMII					RSVD	RSVD	VDDCX_1	VSSX_0	M
N	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	GPIO_00					VSSX_0	VDDPX_3	RSVD	RSVD	N
P	VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	GPIO_01					RSVD	VDDCX_1	RSVD	RSVD	P
	15	16	17	18	19	20	21	22	23	24	25	26	27	28	

EBI1* PCI* USB* | GMII* | GPIO* | VDDPX* | VDDCX* | VDD_APC* | *VDD* | *VSS* | VSSX_0

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Table 2-5 IPQ8068 BGA519 Pin assignments (lower left quadrant)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
R	EBI1_CK0B	EBI1_CK0	VSSX_0	VSSX_0					VDDA_CDC_EBI1_CA	VSSX_0	VDDCX_1	VDD_HF_PLL12	VDD_HF_PLL12	VSSX_0	R
T	EBI1_WEN	EBI1_CKE0	EBI1_CS1N	VDDPX_1					VSSX_0	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	T
U	EBI1_CS0N	EBI1_RESETN	EBI1_ODT0	EBI1_CKE1					VSSX_0	VSSX_0	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	U
V	EBI1_CASN	EBI1_RASN	EBI1_DQ08	VREF_EBI1_0					VDDCX_1	VSSX_0	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	V
W	EBI1_DQ09	EBI1_DQ10	EBI1_DQ11	VDDA_CDC_EBI1_0					VDD_HF_PLL9	VSS_HF_PLL9	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	W
Y	EBI1_DQ12	EBI1_DQ13	EBI1_DQ15	VSSX_0					VDDCX_1	VDDCX_1	VDD_APC0	VDD_APC0	VSSX_0	VDD_APC1	Y
AA	EBI1_DQSB1	EBI1_DQS1	EBI1_DM1	VDDPX_1											AA
AB	EBI1_DQ14	EBI1_DM0	EBI1_DQ00	EBI1_DQ01											AB
AC	EBI1_DQS0	EBI1_DQSB0	EBI1_DQ03	EBI1_DQ05											AC
AD	EBI1_DQ02	EBI1_DQ04	VDDPX_1	EBI1_DQ07											AD
AE	EBI1_DQ06	TRST_N	VDDPX_1		VSS_SENSE_APC_0_1	VDD_SENSE_APC_0	VDD_SENSE_APC_1	VDDPX_3	VSSX_0	GPIO_43	GPIO_47	VDDPX_3	VSSX_0	VDD_CDC_SDC3	AE
AF	RTCK	TMS	VSSX_0	VDDPX_3	VDDPX_3	TDO	MODE_0	GPIO_37	GPIO_34	GPIO_44	GPIO_46	SDC3_DATA_1	VDDPX_VBIAS_SDC	VDD_CDC_SDC1	AF
AG	VSSX_0	VSSX_0	TDI	TCK	MODE_1	GPIO_26	GPIO_40	GPIO_36	GPIO_42	GPIO_45	SDC3_CMD	SDC3_CLK	SDC3_DATA_3	VDDPX_2	AG
AH		VSSX_0	RESIN_N	RESOUT_N	SRST_N	GPIO_35	GPIO_39	GPIO_41	GPIO_38	SDC3_DATA_0	SDC3_DATA_2	GPIO_25	GPIO_24	GPIO_23	AH
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

EBI1* PCI* USB* | GMII* | GPIO* | VDDPX* | VDDCX* | VDD_APC* | *VDD* | *VSS* | VSSX_0

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Table 2-6 IPQ8068 BGA519 Pin assignments (lower right quadrant)

	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
R	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_ USB3_SS	VDDD_ USB3_SS					VDDD_ USB3_ SS_VPTX	USB3_ SS_ RESREF	VSSA_ USB3_SS	VSSA_ USB3_SS	R
T	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0	VSSA_ USB3_HS	VDDD_ USB3_HS					VDDA_ USB3_ SS_1P8	USB3_ HS_VBUS	USB3_ SS_TX0M	USB3_ SS_TX0P	T
U	VDD_ APC1	VSSX_0	VDDCX_1	VDDCX_1	VSSX_0	VSSX_0					VDDA_ USB3_ HS_1P8	VDDA_ USB3_ HS_3P3	USB3_ SS_RX0M	USB3_ SS_RX0P	U
V	VDD_ APC1	VSSX_0	VDDCX_1	VDDCX_1	VSSA_ USB1_SS	VDDD_ USB1_SS					VSSA_ USB3_HS	USB3_ HS_ TXRTUNE	USB3_ HS_DM	USB3_ HS_DP	V
W	VDD_ APC1	VSSX_0	VDD_ QFPROM _BLOW	VDDCX_1	VSSA_ USB1_HS	VDDD_ USB1_HS					USB3_ HS_ SYSCLOC K	USB3_ HS_ID	USB1_3_ SS_REF_ CLKP	USB1_3_ SS_REF_ CLKM	W
Y	VDD_ APC1	VSS_HF_ PLL10	VSS_ SR2_PLL	VSS_SR_ PLL0	VSSX_0	VSSX_0					VDDD_ USB1_ SS_VPTX	VSSA_ USB1_SS	USB1_ SS_TX0M	USB1_ SS_TX0P	Y
AA											USB1_ SS_ RESREF	VDDA_ USB1_ SS_1P8	USB1_ SS_RX0M	USB1_ SS_RX0P	AA
AB											VSSA_ USB1_HS	VDDA_ USB1_ HS_1P8	USB1_ HS_VBUS	USB1_ HS_ SYSCLOC K	AB
AC											USB1_ HS_ TXRTUNE	USB1_ HS_ID	USB1_ HS_DM	USB1_ HS_DP	AC
AD											VDDA_ USB1_ HS_3P3	VDDPX_3	VSSX_0	VSSX_0	AD
AE	VSSX_0	VDDPX_3	VSSX_0	VDD_ HF_ PLL10	VDD_ SR2_PLL	VDD_SR_ PLL	VDDPX_5	VSSX_0	VDDPX_3	GPIO_10		VDDPX_3	GPIO_48	GPIO_49	AE
AF	VSSX_0	VDDPX_3	GPIO_12	NC	NC	SLEEP_ CLK	GPIO_65	GPIO_63	VDDPX_ VBIAS_ UIM	VDDPX_4	VDDPX_4	VSSX_0	GPIO_50	GPIO_54	AF
AG	VSSX_0	GPIO_04	GPIO_13	NC	SLEEP_ CLK_ 32K_OUT	NC	CXO	GPIO_64	GPIO_68	GPIO_06	GPIO_08	GPIO_53	VSSX_0	VSSX_0	AG
AH	GPIO_22	GPIO_05	GPIO_03	GPIO_11	CLK_ 25M_ OUT	NC	NC	NC	GPIO_67	USB2_ HSIC_ CAL	GPIO_09	GPIO_07	VSSX_0		AH
	15	16	17	18	19	20	21	22	23	24	25	26	27	28	

EBI1* PCI* USB* | GMII* | GPIO* | VDDPX* | VDDCX* | VDD_ APC* | *VDD* | *VSS* | VSSX_0

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2.3 Pin descriptions

Descriptions of pins are presented in the following sections.

2.3.1 Power supply pins

Table 2-7 Power supply pins

Pin #	Pin name	Voltage	Functional description
Pad Power			
E3, H4, N4, T4, AA4, AD3, AE3	VDDPX_1	1.35	EBI1
AG14	VDDPX_2	1.8/2.95	SDC3, dual voltage
C25, D9, E25, E26, N26, P9, AD26, AE8, AE12, AE16, AE23, AE26, AF4, AF5, AF16	VDDPX_3	1.8	General power: MODE, JTAG, I2S, SDC1, GPIOs
AF24, AF25	VDDPX_4	1.2/1.8	HSIC, dual voltage
AE21	VDDPX_5	1.8/2.95	UIM1, dual voltage
C4, C5	VDDPX_6	2.5	RGMII
QSGMII PHY			
E27	VDDA_QSGMII_DRV	1.1	Analog power for clock driver 0 & 1 & 2
K26	VDDA_QSGMII_H18	1.8	Analog high-voltage power for bandgap and bias generation
D27, J26	VDDA_QSGMII_PLL	1.1	Analog power for PLL
H26	VDDA_QSGMII_RX	1.1	Analog power for QSGMII/SGMII0 RX
G26	VDDA_QSGMII_TX	1.1	Analog power for QSGMII/SGMII0 TX
L20, L26	VDDA_SGMII	1.1	Analog power for SGMII1 and SGMII2
SATA			
M27	VDDA_SATA	1.1	Transmit supply
P26	VDDA_SATA_VP	1.1	Analog and digital supply
N26	VDDA_SATA_VPH	1.8	High-voltage power
USB1			
AB26	VDDA_USB1_HS_1P8	1.8	High-speed high-voltage
AD25	VDDA_USB1_HS_3P3	3.3	High-speed high-voltage
W20	VDDD_USB1_HS	1.1	High-speed analog and digital supply
AA26	VDDA_USB1_SS_1P8	1.8	Super-speed high-voltage
V20	VDDD_USB1_SS	1.1	Super-speed analog and digital supply
Y25	VDDD_USB1_SS_VPTX	1.1	Super-speed PHY transmit supply
USB3			
U25	VDDA_USB3_HS_1P8	1.8	High-speed high-voltage
U26	VDDA_USB3_HS_3P3	3.3	High-speed high-voltage
T20	VDDD_USB3_HS	1.1	High-speed analog and digital supply
T25	VDDA_USB3_SS_1P8	1.8	Super-speed high-voltage
R20	VDDD_USB3_SS	1.1	Super-speed analog and digital supply
R25	VDDD_USB3_SS_VPTX	1.1	Super-speed PHY transmit supply

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Table 2-7 Power supply pins (cont.)

Pin #	Pin name	Voltage	Functional description
PCIe			
J18	VDDA_VPH_PCIE1	1.8	High voltage supply
J17	VDDA_VPH_PCIE2	1.8	
J16	VDDA_VPH_PCIE3	1.8	
C18	VDDA_VPTX0_PCIE1	1.1	Transmit supply
C17	VDDA_VPTX0_PCIE2	1.1	
C15	VDDA_VPTX0_PCIE3	1.1	
D18	VDDA_VP_PCIE1	1.1	Analog and digital supply
D17	VDDA_VP_PCIE2	1.1	
C14	VDDA_VP_PCIE3	1.1	
Bias, Calibration, Reference, PLLs			
D7	VDDPX_VBIAS_RGMII	0.85	Provided by resistor divider. Should be capable of sinking current.
AF13	VDDPX_VBIAS_SDC	1.25	
AF23	VDDPX_VBIAS_UIM	1.25	
K4	VDDA_CDC_EB1_1	1.1	Calibration delay circuit
W4	VDDA_CDC_EBI1_0	1.1	Calibration delay circuit
R9	VDDA_CDC_EBI1_CA	1.1	Calibration delay circuit
AF14	VDD_CDC_SDC1	1.1	Calibration delay circuit
AE14	VDD_CDC_SDC3	1.1	Calibration delay circuit
W17	VDD_QFPROM	1.8	Power for programming Q-fuses; otherwise connect to ground
AE6	VDD_SENSE_APC_0	–	Sense node for Krait application microprocessor 0 supply voltage
AE7	VDD_SENSE_APC_1	–	Sense node for Krait application microprocessor 1 supply voltage
V4	VREF_EBI1_0	1.35/2	Provided by resistor divider.
L4	VREF_EBI1_1	1.35/2	
AE18	VDD_HF_PLL10	1.8	Power for PLL circuits
R12, R13	VDD_HF_PLL12	1.8	
W9	VDD_HF_PLL9	1.8	
AE19	VDD_SR2_PLL	1.8	
AE20	VDD_SR_PLL	1.1	

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Table 2-7 Power supply pins (cont.)

Pin #	Pin name	Voltage	Functional description
Core Voltages			
J9, J14, J19, K9, K14, K18, K19, L15, L16, M11, M12, M15, M16, N9, N10, N13, N14, N17, N18, P10, P14, P17, P18, R11, R15, R16, T11, T12, T15, T16, U17, U18, V9, V17, V18, W18, Y9, Y10	VDDCX_1	1.1	Digital core
U11, U12, V11, V12, W11, W12, Y11, Y12	VDD_APC0	1.1	Application processor core
U14, U15, V14, V15, W14, W15, Y14, Y15	VDD_APC1	1.1	
J11, J12, K11, K12, L11, L12	VDD_UBI32	1.1	NSS processor core

2.3.2 Ground pins

Table 2-8 Ground pins

Pin #	Pin Name	Description
A2, A27, B1, B2, B27, B28, C12, C13, C16, C19, C27, D3, D8, D12, D13, D15, D20, E28, F25, F26, J4, J10, J13, J15, J20, J25, K10, K13, K15, K16, K17, K20, L9, L10, L13, L14, L17, L18, L19, L25, M9, M10, M13, M14, M17, M18, M19, M20, M28, N11, N12, N15, N16, N19, N25, P3, P4, P11, P12, P13, P15, P16, P19, R3, R4, R10, R14, R17, R18, R19, R27, R28, T9, T10, T13, T14, T17, T18, T19, U9, U10, U13, U16, U19, U20, V10, V13, V16, V19, V25, W10, W13, W16, W19, Y4, Y13, Y16, Y17, Y18, Y19, Y20, Y26, AB25, AD27, AD28, AE9, AE13, AE15, AE17, AE22, AF3, AF15, AF26, AG1, AG2, AG15, AG27, AG28, AH2, AH27	GND	Ground pins shorted together in package to VSSX_0.
AE5	VSS_SENSE_APC_0_1	Separate ground pin for power management sensing.

2.3.3 No-connect and reserved pins

Table 2-9 No-connect and reserved pins

Pin #	Pin Name	Description
AF18, AF19, AG18, AG20, AH20, AH21, AH22	NC, RSVD	Reserved or do not connect.

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2.3.4 Clocks and reset

Table 2-10 Clocks and Reset

Pin #	Pin name	Voltage	Type	Functional description
AH19	CLK_25M_OUT	1.8	O	Free-running 25 MHz clock output. Can be looped back on board to CXO pin.
AG21	CXO	1.8	AI:IE	CXO clock input at 25 MHz
AH3	RESIN_N	1.8	IS:hvin	Hardware reset input to system
AH4	RESOUT_N	1.8	O	Reset output generated by synchronized RESIN_N and by wdog_ reset.
AF20	SLEEP_CLK	1.8	IS:hvin	Digital sleep clock at 32 KHz
AG19	SLEEP_CLK_32K_OUT	1.8	O	32 KHz digital sleep clock output. Can be looped back on board to SLEEP_CLK.

The rising edge of RESOUT_N is delayed from the rising edge of RESIN_N by a few microprocessor clocks.

2.3.5 Memory support functions

Table 2-11 PCDDR3 External Bus Interface Channel 0 (2 CS)

Pin #	Pin name	Voltage	Type	Functional description
M1	EBI1_A00	P1	O BS:IE	Command/Address
M3	EBI1_A01	P1	O BS:IE	Command/Address
L1	EBI1_A02	P1	O BS:IE	Command/Address
L2	EBI1_A03	P1	O BS:IE	Command/Address
P2	EBI1_A04	P1	O BS:IE	Command/Address
J1	EBI1_A05	P1	O BS:IE	Command/Address
J2	EBI1_A06	P1	O BS:IE	Command/Address
J3	EBI1_A07	P1	O BS:IE	Command/Address
H2	EBI1_A08	P1	O BS:IE	Command/Address
H1	EBI1_A09	P1	O BS:IE	Command/Address
L3	EBI1_A10	P1	O BS:IE	Clock enable
K3	EBI1_A11	P1	O BS:IE	Clock enable
K1	EBI1_A12	P1	O BS:IE	Chip select
K2	EBI1_A13	P1	O BS:IE	Chip select
M2	EBI1_A14	P1	O BS:IE	Command/Address
N2	EBI1_BA0	P1	O BS:IE	Byte Access

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Table 2-11 PCDDR3 External Bus Interface Channel 0 (2 CS) (cont.)

Pin #	Pin name	Voltage	Type	Functional description
N1	EBI1_BA1	P1	O BS:IE	Byte Access
N3	EBI1_BA2	P1	O BS:IE	Byte Access
P1	EBI1_CAL	P1	AIO	IO Calibration pad (240 ohm 1%)
V1	EBI1_CASN	P1	O BS:IE	Chip select
R2	EBI1_CK0	P1	O BS:IE	Differential clock (+)
R1	EBI1_CK0B	P1	O BS:IE	Differential clock (-)
T2	EBI1_CKE0	P1	O BS:IE	Clock enable
U4	EBI1_CKE1	P1	O BS:IE	Clock enable
U1	EBI1_CS0N	P1	O BS:IE	Chip select
T3	EBI1_CS1N	P1	O BS:IE	Chip select
AB2	EBI1_DM0	P1	O BS:IE	Data mask
AA3	EBI1_DM1	P1	O BS:IE	Data mask
C1	EBI1_DM2	P1	O BS:IE	Data mask
F4	EBI1_DM3	P1	O BS:IE	Data mask
AB3	EBI1_DQ00	P1	BS:IE	Data
AB4	EBI1_DQ01	P1	BS:IE	Data
AD1	EBI1_DQ02	P1	BS:IE	Data
AC3	EBI1_DQ03	P1	BS:IE	Data
AD2	EBI1_DQ04	P1	BS:IE	Data
AC4	EBI1_DQ05	P1	BS:IE	Data
AE1	EBI1_DQ06	P1	BS:IE	Data
AD4	EBI1_DQ07	P1	BS:IE	Data
V3	EBI1_DQ08	P1	BS:IE	Data
W1	EBI1_DQ09	P1	BS:IE	Data
W2	EBI1_DQ10	P1	BS:IE	Data
W3	EBI1_DQ11	P1	BS:IE	Data
Y1	EBI1_DQ12	P1	BS:IE	Data
Y2	EBI1_DQ13	P1	BS:IE	Data
AB1	EBI1_DQ14	P1	BS:IE	Data
Y3	EBI1_DQ15	P1	BS:IE	Data
F3	EBI1_DQ16	P1	BS:IE	Data
G4	EBI1_DQ17	P1	BS:IE	Data
F2	EBI1_DQ18	P1	BS:IE	Data
F1	EBI1_DQ19	P1	BS:IE	Data
G3	EBI1_DQ20	P1	BS:IE	Data
G2	EBI1_DQ21	P1	BS:IE	Data

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Table 2-11 PCDDR3 External Bus Interface Channel 0 (2 CS) (cont.)

Pin #	Pin name	Voltage	Type	Functional description
H3	EBI1_DQ22	P1	BS:IE	Data
G1	EBI1_DQ23	P1	BS:IE	Data
A3	EBI1_DQ24	P1	BS:IE	Data
A4	EBI1_DQ25	P1	BS:IE	Data
B4	EBI1_DQ26	P1	BS:IE	Data
D6	EBI1_DQ27	P1	BS:IE	Data
E4	EBI1_DQ28	P1	BS:IE	Data
B3	EBI1_DQ29	P1	BS:IE	Data
D5	EBI1_DQ30	P1	BS:IE	Data
C2	EBI1_DQ31	P1	BS:IE	Data
AC1	EBI1_DQS0	P1	BS:IE	Data strobe
AA2	EBI1_DQS1	P1	BS:IE	Data strobe
E2	EBI1_DQS2	P1	BS:IE	Data strobe
D2	EBI1_DQS3	P1	BS:IE	Data strobe
AC2	EBI1_DQSB0	P1	BS:IE	Data strobe
AA1	EBI1_DQSB1	P1	BS:IE	Data strobe
E1	EBI1_DQSB2	P1	BS:IE	Data strobe
D1	EBI1_DQSB3	P1	BS:IE	Data strobe
U3	EBI1_ODT0	P1	O BS:IE	ODT
M4	EBI1_ODT1	P1	O BS:IE	ODT
V2	EBI1_RASN	P1	O BS:IE	Chip select
U2	EBI1_RESETN	P1	O BS:IE	Reset
T1	EBI1_WEN	P1	O BS:IE	Chip select

2.3.6 PCI Express (PCIe)

A DBDC Wi-Fi connectivity system is connected through two PCIe interfaces. A third PCIe interface is available for a sensor radio or other purposes.

Table 2-12 PCI Express (PCIe)

Pin #	Pin name	Voltage	Type	Functional description
PCIE1				
A18	PCIE1_HSI_0_N	–	AI	PCIe receive lane – minus
B18	PCIE1_HSI_0_P	–	AI	PCIe receive lane – plus
A19	PCIE1_HSO_0_N	–	AO	PCIe transmit lane – minus

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Table 2-12 PCI Express (PCIe) (cont.)

Pin #	Pin name	Voltage	Type	Functional description
B19	PCIE1_HSO_0_P	–	AO	PCIe transmit lane – plus
D19	PCIE1_REXT	–	AIO	PCIe external reference resistor (200 ohm 1%)
PCIE2				
A15	PCIE2_HSI_0_N	–	AI	PCIe receive lane – minus
B15	PCIE2_HSI_0_P	–	AI	PCIe receive lane – plus
A16	PCIE2_HSO_0_N	–	AO	PCIe transmit lane – minus
B16	PCIE2_HSO_0_P	–	AO	PCIe transmit lane – plus
D16	PCIE2_REXT	–	AIO	PCIe external reference resistor (200 ohm 1%)
PCIE3				
A12	PCIE3_HSI_0_N	–	AI	PCIe receive lane – minus
B12	PCIE3_HSI_0_P	–	AI	PCIe receive lane – plus
A13	PCIE3_HSO_0_N	–	AO	PCIe transmit lane – minus
B13	PCIE3_HSO_0_P	–	AO	PCIe transmit lane – plus
D14	PCIE3_REXT	–	AIO	PCIe external reference resistor (200 ohm 1%)

The pins shown in Table 2-13 are available for testing. However, in normal operation, REF_CLK to the PCIe PHY is provided internally, and these pins should be left unconnected.

Table 2-13 PCIe test clocks

Pin #	Pin name	Voltage	Type	Functional description
B20	PCIE1_REFCLK_N	–	AI	Reference clock to PCIe PHY – minus
A20	PCIE1_REFCLK_P	–	AI	Reference clock to PCIe PHY – plus
B17	PCIE2_REFCLK_N	–	AI	Reference clock to PCIe PHY – minus
A17	PCIE2_REFCLK_P	–	AI	Reference clock to PCIe PHY – plus
B14	PCIE3_REFCLK_N	–	AI	Reference clock to PCIe PHY – minus
A14	PCIE3_REFCLK_P	–	AI	Reference clock to PCIe PHY – plus

Table 2-14 PCIe Clocking

Pin #	Pin name	Voltage	Type	Functional description
A23	PCIE1_CLK_OUT_N	–	AO	Clock to PCIe end point.
B23	PCIE1_CLK_OUT_P	–	AO	Clock to PCIe end point.
A21	PCIE2_CLK_OUT_N	–	AO	Clock to PCIe end point.
B21	PCIE2_CLK_OUT_P	–	AO	Clock to PCIe end point.

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Table 2-14 PCIe Clocking (cont.)

Pin #	Pin name	Voltage	Type	Functional description
A22	PCIE3_CLK_OUT_N	–	AO	Clock to PCIe end point.
B22	PCIE3_CLK_OUT_P	–	AO	Clock to PCIe end point.

In Table 2-15, pins labeled with the suffix "_A" and pins labeled with the suffix "_B" are alternate mappings of the same PCIE1 unit's signals.

Table 2-15 PCIe hot-plug signals and GPIO

Pin #	Pin name	Alt function	Voltage	Type	Functional description
PCIE1					
AH16	PCIE1_CLK_REQ_N	GPIO_05	P3	I	Clock request from endpoint
AH18	PCIE1_PRSNT_2_N_A	GPIO_11	P3	I	Power presence detection
AH17	PCIE1_PRSNT_2_N_B	GPIO_03	P3	I	
AF17	PCIE1_PWREN_A	GPIO_12	P3	O	Signal to power up/down the transceiver
	PCIE1_PWREN_N_A	GPIO_12	P3	O	
AG16	PCIE1_PWREN_B	GPIO_04	P3	O	
	PCIE1_PWREN_N_B	GPIO_04	P3	O	
AG17	PCIE1_PWRFLT_N_A	GPIO_13	P3	I	Power fault indication
AH16	PCIE1_PWRFLT_N_B	GPIO_05	P3	I	
AH17	PCIE1_RST_N	GPIO_03	P3	O	Reset
AG16	PCIE1_WAKE_N	GPIO_04	P3	I	Request from endpoint to awaken host from low-power state
PCIE2					
AF27	PCIE2_CLK_REQ	GPIO_50	P3	I	Clock request from endpoint
AH18	PCIE2_PRSNT_2_N_A	GPIO_11	P3	I	Power presence detection
AE27	PCIE2_PRSNT_2_N_B	GPIO_48	P3	I	
AF17	PCIE2_PWREN_A	GPIO_12	P3	O	Signal to power up/down the transceiver
AE28	PCIE2_PWREN_B	GPIO_49	P3	O	
AF17	PCIE2_PWREN_N_A	GPIO_12	P3	O	
AE28	PCIE2_PWREN_N_B	GPIO_49	P3	O	
AG17	PCIE2_PWRFLT_N_A	GPIO_13	P3	I	Power fault indication
AF27	PCIE2_PWRFLT_N_B	GPIO_50	P3	I	
AE27	PCIE2_RST_N	GPIO_48	P3	O	Reset
AE28	PCIE2_WAKE_N	GPIO_49	P3	I	Request from endpoint to awaken host from low-power state
PCIE3					
AF21	PCIE3_CLK_REQ	GPIO_65	P3	I	Clock request from endpoint
AH18	PCIE3_PRSNT_2_N	GPIO_11	P3	I	Power presence detection

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Table 2-15 PCIe hot-plug signals and GPIO (cont.)

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AF17	PCIE3_PWREN	GPIO_12	P3	O	PCIe signal to power up/down the transceiver
	PCIE3_PWREN_N	GPIO_12	P3	O	
AG17	PCIE3_PWRFLT_N	GPIO_13	P3	I	Power fault indication
AF22	PCIE3_RST_N	GPIO_63	P3	O	Reset
AG22	PCIE3_WAKE_N	GPIO_64	P3	I	Request from endpoint to awaken host from low-power state

2.3.7 JTAG interface

Table 2-16 Primary JTAG Interface

Pin #	Pin name	Voltage	Type	Functional description
AF1	RTCK	P3	DO	JTAG return test clock
AH5	SRST_N	P3	DI	JTAG reset for Debug
AG4	TCK/SWD_CLK	P3	DI	JTAG test clock input, SWD clock input
AG3	TDI	P3	DI	JTAG test data input
AF6	TDO	P3	DO-Z	JTAG test data output
AF2	TMS/SWD_DATA	P3	DI	JTAG test mode state, SWD data input
AE2	TRST_N	P3	DI	JTAG test reset

2.3.8 QSGMII, SGMII

The four GMAC controllers can operate in any of the configurations shown in Table 2-17.

Table 2-17 GMAC configurations

Configuration	GMAC 1	GMAC 2	GMAC 3	GMAC 4
A	QSGMII			
B	RGMII (2.5 V)	SGMII (SGMII0)	SGMII (SGMII1)	SGMII (SGMII2)
C	RGMII (2.5 V)	RGMII (1.8 V)	SGMII (SGMII1)	SGMII (SGMII2)

Table 2-18 QSGMII, SGMII

Pin #	Pin name	Voltage	Type	Functional description
K25	QSGMII_CLK25M_OUT	1.1	AO	A clean 25MHz clock output. Can be used as a reference clock for external PHY chips.
G27	QSGMII_QSI0_N	1.1	AI	QSGMII 5 Gbps differential negative input, or SGMII0 1.25 Gbps differential negative input
G28	QSGMII_QSI0_P	1.1	AI	QSGMII 5 Gbps differential positive input, or SGMII0 1.25 Gbps differential positive input

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Table 2-18 QSGMII, SGMII (cont.)

Pin #	Pin name	Voltage	Type	Functional description
F27	QSGMII_QSO0_N	1.1	AO	QSGMII 5 Gbps differential negative output, or SGMII0 1.25 Gbps differential negative output
F28	QSGMII_QSO0_P	1.1	AO	QSGMII 5 Gbps differential positive output, or SGMII0 1.25 Gbps differential negative output
H25	QSGMII_RESREF	–	AI	Bias. An external resistor (1.5 Kohm, ±1%) connected to ground.
G25	QSGMII_SEL_25M	1.1	I PU	Clock source selection signal for QSGMII_XTLI and QSGMII_XTLO. 1 - 25 MHz clock source; 0 - reserved
C28	QSGMII_XTLI	–		Connect 25 MHz crystal oscillator across these pins.
D28	QSGMII_XTLO	–		
J27	SGMII_SI1_N	–	AI	SGMII1 1.25 Gbps differential negative input
J28	SGMII_SI1_P	–	AI	SGMII1 1.25 Gbps differential positive input
K27	SGMII_SI2_N	–	AI	SGMII2 1.25 Gbps differential negative input
K28	SGMII_SI2_P	–	AI	SGMII2 1.25 Gbps differential positive input
H27	SGMII_SO1_N	–	AO	SGMII1 1.25 Gbps differential negative output
H28	SGMII_SO1_P	–	AO	SGMII1 1.25 Gbps differential positive output
L27	SGMII_SO2_N	–	AO	SGMII2 1.25 Gbps differential negative output
L28	SGMII_SO2_P	–	AO	SGMII2 1.25 Gbps differential positive output

2.3.9 RGMII

Two sets of RGMII pins are available:

1. Dedicated (2.5 V).
2. Muxed with GPIOs (1.8 V).

Table 2-19 RGMII (dedicated 2.5 V)

Pin #	Pin name	Voltage	Type	Functional description
A6	RGMII_CK_RX	P6	I PD:nppukp	Receive reference clock.
C9	RGMII_CK_TX	P6	O PD:nppukp	Transmit reference clock.
B5	RGMII_CTL_RX	P6	I PD:nppukp	Receive control.
B7	RGMII_CTL_TX	P6	O PD:nppukp	Transmit control.
A5	RGMII_RX_0	P6	I PD:nppukp	Receive data.
C6	RGMII_RX_1	P6	I PD:nppukp	
B6	RGMII_RX_2	P6	I PD:nppukp	
C7	RGMII_RX_3	P6	I PD:nppukp	

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Table 2-19 RGMII (dedicated 2.5 V) (cont.)

Pin #	Pin name	Voltage	Type	Functional description
A7	RGMII_TX_0	P6	O PD:nppukp	Transmit data.
C8	RGMII_TX_1	P6	O PD:nppukp	
A8	RGMII_TX_2	P6	O PD:nppukp	
B8	RGMII_TX_3	P6	O PD:nppukp	

Table 2-20 RGMII2 (muxed 1.8 V)

Pin #	Pin name	Alt function	Voltage	Type	Functional description
D23	RGMII2_CK_RX	GPIO_51	P3	I PD:nppukp	Receive reference clock.
C23	RGMII2_CK_TX	GPIO_27	P3	O PD:nppukp	Transmit reference clock.
C20	RGMII2_CTL_RX	GPIO_62	P3	I PD:nppukp	Receive control.
C24	RGMII2_CTL_TX	GPIO_32	P3	O PD:nppukp	Transmit control.
C21	RGMII2_RX_0	GPIO_61	P3	I PD:nppukp	Receive data.
D21	RGMII2_RX_1	GPIO_60	P3	I PD:nppukp	
C22	RGMII2_RX_2	GPIO_59	P3	I PD:nppukp	
D22	RGMII2_RX_3	GPIO_52	P3	I PD:nppukp	
A24	RGMII2_TX_0	GPIO_31	P3	O PD:nppukp	Transmit data.
B24	RGMII2_TX_1	GPIO_30	P3	O PD:nppukp	
A25	RGMII2_TX_2	GPIO_29	P3	O PD:nppukp	
B25	RGMII2_TX_3	GPIO_28	P3	O PD:nppukp	

2.3.10 Secure Digital Card (SDC1 and SDC3 Interfaces)

The SDC blocks provide an SD interface for connections to SD-card devices, SDIO devices, and eMMC NAND devices. The Secure Digital Card Controller supports up to SD3.0 specification devices as well as SDIO and MMC, and is fully backward compatible. IPQ8064 has two instances of the SDCC

- SDC1 – 8-bit for eMMC boot/storage.
- SDC3 – 4-bit for external SD/MMC card.

Table 2-21 Secure Digital Card 1 (SDC1) Interface

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AG9	SDC1_CLK	GPIO_42	P3	B	Clock
AG10	SDC1_CMD	GPIO_45	P3	B	Command
AF10	SDC1_DATA[0]	GPIO_44	P3	B	Data [0]
AE10	SDC1_DATA[1]	GPIO_43	P3	B	Data [1]

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Table 2-21 Secure Digital Card 1 (SDC1) Interface (cont.)

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AH8	SDC1_DATA[2]	GPIO_41	P3	B	Data [2]
AG7	SDC1_DATA[3]	GPIO_40	P3	B	Data [3]
AE11	SDC1_DATA[4]	GPIO_47	P3	B	Data [4]
AF11	SDC1_DATA[5]	GPIO_46	P3	B	Data [5]
AH7	SDC1_DATA[6]	GPIO_39	P3	B	Data [6]
AH9	SDC1_DATA[7]	GPIO_38	P3	B	Data [7]

Table 2-22 Secure Digital Card 3 (SDC3) Interface

Pin #	Pin name	Voltage	Type	Functional description
AG12	SDC3_CLK	P2	O	Clock.
AG11	SDC3_CMD	P2	B	Command.
AH10	SDC3_DATA_0	P2	B	Data [0].
AF12	SDC3_DATA_1	P2	B	Data [1].
AH11	SDC3_DATA_2	P2	B	Data [2].
AG13	SDC3_DATA_3	P2	B	Data [3].

2.3.11 Sense Lines

Table 2-23 Sense Lines

Pin #	Pin name	Voltage	Type	Functional description
AE6	VDD_SENSE_APC_0	–	V_A	Sense node for Krait application microprocessor 0 supply voltage
AE7	VDD_SENSE_APC_1	–	V_A	Sense node for Krait application microprocessor 1 supply voltage
AE5	VSS_SENSE_APC_0_1	–	G_A	Sense nodes ground.

2.3.12 Serial ATA (SATA)

Table 2-24 Serial ATA (SATA)

Pin #	Pin name	Voltage	Type	Functional description
P25	SATA_REXT	–	AIO	SATA external reference resistor (200 ohm 1%)
P27	SATA_RXM	–	AI	SATA receive – minus
P28	SATA_RXP	–	AI	SATA receive – plus
N27	SATA_TXM	–	AIO	SATA transmit – minus
N28	SATA_TXP	–	AIO	SATA transmit – plus

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The pins shown in Table 2-25 are available for testing. However, under normal conditions, the reference clock for the SATA PHY is provided internally, and these pins should be left unconnected.

Table 2-25 SATA test clocks

Pin #	Pin name	Voltage	Type	Functional description
M26	SATA_CLKM	–	AI	Reference clock to SATA PHY – minus
M25	SATA_CLKP	–	AI	Reference clock to SATA PHY – plus

2.3.13 Modes

Primary mode selection signals.

Table 2-26 Mode Selection

Pin #	Pin name	Type	Functional description
AF7	MODE_0	PD	Leave unconnected.
AG5	MODE_1		

2.3.14 USB 3.0/2.0 PHY

Table 2-27 USB 3.0/2.0 PHY

Pin #	Pin name	Voltage	Type	Functional description
AC27	USB1_HS_DM	–	AIO	USB1 HS data minus
AC28	USB1_HS_DP	–	AIO	USB1 HS data plus
AC26	USB1_HS_ID	–	AIO	USB1 HS ID (differentiates mini A and B plugs)
AC25	USB1_HS_TXRTUNE	–	AIO	USB1 HS external resistor (200 ohm 1%)
AB27	USB1_HS_VBUS	–	V_A	USB1 HS 5V bus power
AA25	USB1_SS_RESREF	–	AI	USB1 SS external resistor – Kelvin connection (200 ohm 1%)
AA27	USB1_SS_RX0M	–	AI	USB1 SS receive data minus
AA28	USB1_SS_RX0P	–	AI	USB1 SS receive data plus
Y27	USB1_SS_TX0M	–	AO	USB1 SS transmit data minus
Y28	USB1_SS_TX0P	–	AO	USB1 SS transmit data plus
V27	USB3_HS_DM	–	AIO	USB3 HS data minus
V28	USB3_HS_DP	–	AIO	USB3 HS data plus
W26	USB3_HS_ID	–	AIO	USB3 HS ID (differentiates mini A and B plugs)
V26	USB3_HS_TXRTUNE	–	AIO	USB3 HS external resistor (200 ohm 1%)
T26	USB3_HS_VBUS	–	V_A	USB3 HS 5 V supply pin

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Table 2-27 USB 3.0/2.0 PHY (cont.)

Pin #	Pin name	Voltage	Type	Functional description
R26	USB3_SS_RESREF	–	AI	USB3 SS external resistor – Kelvin connection (200 ohm 1%)
U27	USB3_SS_RX0M	–	AI	USB3 SS receive data minus
U28	USB3_SS_RX0P	–	AI	USB3 SS receive data plus
T27	USB3_SS_TX0M	–	AO	USB3 SS transmit data minus
T28	USB3_SS_TX0P	–	AO	USB3 SS transmit data plus

The pins shown in Table 2-28 are available for testing. However, under normal conditions, the 100 MHz and 50 MHz reference clocks to the USB PHY are provided internally, and these pins should be left unconnected.

Table 2-28 USB test clocks

Pin #	Pin name	Voltage	Type	Functional description
W28	USB1_3_SS_REF_CLKM	–	AI	USB1 and USB3 reference clock minus
W27	USB1_3_SS_REF_CLKP	–	AI	USB1 and USB3 reference clock plus
AB28	USB1_HS_SYSCLOCK	–	AI	USB1 HS system clock
W25	USB3_HS_SYSCLOCK	–	AI	USB3 HS system clock

2.3.15 USB HSIC

Table 2-29 USB HSIC

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AH24	USB2_HSIC_CAL	–	P5	B NP:pupdkp	HSIC calibration. Connect to ground through 240 ohm (± 1%) resistor.
AG23	USB2_HSIC_DATA	GPIO_68	P4	B	HSIC data
AH23	USB2_HSIC_STROBE	GPIO_67	P4	B	HSIC strobe

2.3.16 Pulse-code modulation (PCM)

A pulse-code modulation (PCM) interface allows the connection of RJ11 ports for plain-old-telephone-service (POTS) phone systems throught an external SLIC chip. A VoIP software stack on the Krait CPU system can establish communication to the central server to offer voice services.

Table 2-30 PCM

Pin #	Pin name	Alt function	Voltage	Type	Functional description
A10	AUDIO_PCM_CLK	GPIO_17	P3	B	Audio PCM clock
C11	AUDIO_PCM_DIN	GPIO_15	P3	I	Audio PCM data input
B11	AUDIO_PCM_DOUT	GPIO_14	P3	Z	Audio PCM data output
B10	AUDIO_PCM_SYNC	GPIO_16	P3	B	Audio PCM sync

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2.3.17 Boot configuration

Several GPIO signals can be used to configure the secure boot feature and boot device. They are sampled only during power-on reset. Table 2-31 shows the boot configuration signals.

Table 2-31 Boot configuration

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AE27	BOOT_CONFIG[0]	GPIO_48	P3	I	FAST_BOOT configuration. Refer to Table 2-32
AG8	BOOT_CONFIG[1]	GPIO_36	P3	I	
AH6	BOOT_CONFIG[2]	GPIO_35	P3	I	
AF9	BOOT_CONFIG[3]	GPIO_34	P3	I	
D24	BOOT_CONFIG[4]	GPIO_33	P3	I	TCXO frequency selection: 0 - reserved 1 - 25 MHz
AH9	BOOT_CONFIG[5]	GPIO_38	P3	I	Boot type: 1 - FAST_BOOT
AF8	BOOT_CONFIG[6]	GPIO_37	P3	I	BOOT_CONFIG[5:0] usage: 1 - FAST_BOOT boot device configuration
B11	BOOT_FROM_ROM	GPIO_14	P3	I	Primary boot device type: 0 - Primary boot from internal memory 1 - Primary boot from ROM
AH17	WDOG_DISABLE	GPIO_03	P3	I	Watchdog timer disable

Table 2-32 Fast boot device selection

BOOT_CONFIG[3:0]	Mapping
0x01	Boot from SDC1 then SDC3
0x03	Boot from SDC1
0x09	Boot from SPI on GSBI5
0x0A	Boot from SPI on GSBI6
0x0B	Boot from SPI on GSBI7
0x0C	Boot from NAND
0x0D	Reserved
0x0E	Boot from SDC1 in 4-bit mode

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2.3.18 EBI2 NAND flash interface

Table 2-33 EBI2

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AG8	EBI2_ALE_N	GPIO_36	P3	O	Address latch enable
AE11	EBI2_A_D[0]	GPIO_47	P3	B	Multiplexed address or data [7:0]
AF11	EBI2_A_D[1]	GPIO_46	P3	B	
AG10	EBI2_A_D[2]	GPIO_45	P3	B	
AF10	EBI2_A_D[3]	GPIO_44	P3	B	
AE10	EBI2_A_D[4]	GPIO_43	P3	B	
AG9	EBI2_A_D[5]	GPIO_42	P3	B	
AH8	EBI2_A_D[6]	GPIO_41	P3	B	
AG7	EBI2_A_D[7]	GPIO_40	P3	B	
AH7	EBI2_BUSY_N	GPIO_39	P3	I	NAND busy
AH6	EBI2_CLE_N	GPIO_35	P3	O	Command latch enable
AF9	EBI2_NAND_CS_N	GPIO_34	P3	O	NAND chip select
AH9	EBI2_OE_N	GPIO_38	P3	O	Output enable
AF8	EBI2_WE_N	GPIO_37	P3	O	Write enable

2.3.19 General Serial Bus Interface (GSBI)

Thirty-three of the GPIO pins can be configured as up to six General Serial Bus Interfaces.

Each General Serial Bus Interface (GSBI) can be configured for the serial protocols UART, UIM, I^2C, and SPI. At any given time, one of the serial protocols may be selected for use. Also, there are two concurrency schemes that allow for a two-wire UART and I^2C or UIM and I^2C to be used simultaneously. Table 2-34 shows the possible configurations for each GSBI interface.

Table 2-34 GSBI Configurations

Pin	UART	UIM	I^2C	I^2C + UIM	I^2C + 2-wire UART	SPI
3	uart_tx_data	uim_data	gnd_tie	uim_data	uart_tx_data	spi_mosi_data
2	uart_rx_data	uim_clk	gnd_tie	uim_clk	uart_rx_data	spi_miso_data
1	uart_cts_n	unused	i2c_data	i2c_data	i2c_data	spi_cs_n
0	uart_rfr_n	unused	i2c_clk	i2c_clk	i2c_clk	spi_clk
Pin numbers are relative to each of the GSBI interfaces shown in Table 2-35.						

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Table 2-35 General Serial Bus Interfaces

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AF28	GSBI1[0]	GPIO_54	P3	B	Refer to Table 2-34.
AG26	GSBI1[1]	GPIO_53	P3	B	
D22	GSBI1[2]	GPIO_52	P3	B	
D23	GSBI1[3]	GPIO_51	P3	B	
AH12	GSBI2[0]	GPIO_25	P3	B	
AH13	GSBI2[1]	GPIO_24	P3	B	
AH14	GSBI2[2]	GPIO_23	P3	B	
AH15	GSBI2[3]	GPIO_22	P3	B	
AG17	GSBI4[0]	GPIO_13	P3	B	
AF17	GSBI4[1]	GPIO_12	P3	B	
AH18	GSBI4[2]	GPIO_11	P3	B	
AE24	GSBI4[3]	GPIO_10	P3	B	
D10	GSBI5[0]	GPIO_21	P3	B	
A9	GSBI5[1]	GPIO_20	P3	B	
C10	GSBI5[2]	GPIO_19	P3	B	
D11	GSBI5[3]	GPIO_18	P3	B	
AG24	GSBI5_SPI_CS1A_N	GPIO_06	P5	Z	Slave select 1; alternate pin mappings.
C21	GSBI5_SPI_CS1B_N	GPIO_61	P3	Z	
AH26	GSBI5_SPI_CS2A_N	GPIO_07	P5	Z	Slave select 2; alternate pin mappings.
C20	GSBI5_SPI_CS2B_N	GPIO_62	P3	Z	
B9	GSBI5_SPI_CS3_N	GPIO_02	P3	Z	Slave select 3
B24	GSBI6_A[0]	GPIO_30	P3	B	GSBI6 alternate pin mapping A. Refer to Table 2-34.
A25	GSBI6_A[1]	GPIO_29	P3	B	
B25	GSBI6_A[2]	GPIO_28	P3	B	
C23	GSBI6_A[3]	GPIO_27	P3	B	
D26	GSBI6_B[0]	GPIO_58	P3	B	GSBI6 alternate pin mapping B. Refer to Table 2-34.
C26	GSBI6_B[1]	GPIO_57	P3	B	
B26	GSBI6_B[2]	GPIO_56	P3	B	
A26	GSBI6_B[3]	GPIO_55	P3	B	
AH25	GSBI7[0]	GPIO_09	P3	B	Refer to Table 2-34.
AG25	GSBI7[1]	GPIO_08	P5	B	
AH26	GSBI7[2]	GPIO_07	P5	B	
AG24	GSBI7[3]	GPIO_06	P5	B	

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2.3.20 Multiple I2S interface (MI2S)

Some of MI2S unit's signals have alternate mappings: either to pins labeled with the suffix "_A" or to pins labeled with the suffix "_B".

Table 2-36 MI2S

Pin #	Pin name	Alt function	Voltage	Type	Functional description
C24	MI2S_DATA_SD0_A	GPIO_32	P3	B	Serial data channel 0 (alternate pin mappings).
D26	MI2S_DATA_SD0_B	GPIO_58	P3	B	
A24	MI2S_DATA_SD1	GPIO_31	P3	B	Serial data channel 1.
B24	MI2S_DATA_SD2	GPIO_30	P3	B	Serial data channel 2.
D24	MI2S_DATA_SD3	GPIO_33	P3	B	Serial data channel 3.
A25	MI2S_MCLK_A	GPIO_29	P3	O	Master clock (alternate pin mappings).
C26	MI2S_MCLK_B	GPIO_57	P3	O	
B25	MI2S_SCLK_A	GPIO_28	P3	B	Bit clock (alternate pin mappings).
B26	MI2S_SCLK_B	GPIO_56	P3	B	
C23	MI2S_WS_A	GPIO_27	P3	B	Word select (alternate pin mappings).
A26	MI2S_WS_B	GPIO_55	P3	B	

2.3.21 SPDIF

In Table 2-37, pins labeled with the suffix "_A" and pins labeled with the suffix "_B" are alternate mappings of the same SPDIF unit's signals.

Table 2-37 SPDIF

Pin #	Pin name	Alt function	Voltage	Type	Functional description
AE27	SPDIF_TX_A	GPIO_48	P3	O	SPDIF transmit
AE24	SPDIF_TX_B	GPIO_10	P3	O	SPDIF transmit

2.3.22 Transport Stream Interface

The transport stream interface (TSIF) allows over-the-air digital video TV signals to be stored and distributed over the LAN. An MPEG2-TS packet processor is able to perform PID filtering at line speed to extract the user selected elementary stream.

The TSIF block contains dual TSIF cores, one Transport Stream Packet Processor (TSPP) and one BAM core. It supports the following features:

1. Twelve segment ISDBT
2. Up to two physical TSIF interfaces
3. Provide internal DMA capability
4. Packet ID filtering

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5. Transport stream packet processing and packetized elementary stream assembly and header analysis
6. Transport stream packet header analysis and packet adaptation field analysis
7. Time stamp handling and performance counter generation

The TSPP receives TSIF packets from the TSIF core and creates ES packets, which are passed to the software. TSPP block supports TSP Raw processing and TSP PES processing. TSP processing depends on TSP PID and software configuration.

Table 2-38 Transport Stream Interfaces

Pin #	Pin name	Alt function	Voltage	Type	Functional description
A26	TSIF1_CLK	GPIO_55	P3	I	Clock
C26	TSIF1_DATA	GPIO_57	P3	I	Data
B26	TSIF1_EN	GPIO_56	P3	I	Enable
D26	TSIF1_SYNC	GPIO_58	P3	I	Sync
C22	TSIF2_CLK	GPIO_59	P3	I	Clock
C21	TSIF2_DATA	GPIO_61	P3	I	Data
D21	TSIF2_EN	GPIO_60	P3	I	Enable
C20	TSIF2_SYNC	GPIO_62	P3	I	Sync

2.3.23 GPIO Interface

Individual GPIOs are configured by software using GPIO_CFGn registers corresponding to the GPIO number.

Table 2-39 GPIO Interface

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
N20	GPIO_00		P3	B	Configurable I/O
P20	GPIO_01		P3	B	Configurable I/O
B9	GPIO_02		P3	B	Configurable I/O
		GSBI5_SPI_CS3_N		Z	Slave select 3
AH17	GPIO_03		P3	B	Configurable I/O
		PCIE1_RST_N		O	PCIe core reset
		PCIE1_PRSNT_2_N_B		I	PCIe power presence

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
AG16	GPIO_04		P3	B	Configurable I/O
		PCIE1_WAKE_N		I	PCIe link reactivation
		PCIE1_PWREN_N_B		O	PCIe signal to power up/down the transceiver
		PCIE1_PWREN_B		O	
AH16	GPIO_05		P3	B	Configurable I/O
		PCIE1_CLK_REQ_N		I	Clock request from endpoint
		PCIE1_PWRFLT_N_B		I	PCIe power fault indication
AG24	GPIO_06		P5	B	Configurable I/O
		GSBI7_3		B	GSBI7[3]
		USB_FS_OE		B	Output enable
		GSBI5_SPI_CS1A_N		Z	Slave select 1
		USB_FS_OE_N		B	Output enable
AH26	GPIO_07		P5	B	Configurable I/O
		GSBI7_2		B	General Serial Bus Interface
		USB_FS_DAT		B	Data
		GSBI5_SPI_CS2A_N		Z	Slave select 2
AG25	GPIO_08		P5	B	Configurable I/O
		GSBI7_1		B	General Serial Bus Interface
		USB_FS_SE0		B	Single-ended zero
AH25	GPIO_09		P3	B	Configurable I/O
		GSBI7_0		B	General Serial Bus Interface
AE24	GPIO_10		P3	B	Configurable I/O
		GSBI4_3		B	GSBI4[3]
		SPDIF_TX_B		O	SPDIF transmit
AH18	GPIO_11		P3	B	Configurable I/O
		GSBI4_2		B	GSBI4[2]
		PCIE2_PRSNT_2_N_A		I	Power presence detection
		PCIE1_PRSNT_2_N_A		I	
		PCIE3_PRSNT_2_N		I	

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
AF17	GPIO_12		P3	B	Configurable I/O
		GSBI4_1		B	GSBI4[1]
		PCIE2_PWREN_N_A		O	PCIe signal to power up/down the transceiver
		PCIE1_PWREN_N_A		O	
		PCIE3_PWREN_N		O	
		PCIE2_PWREN_A		O	
		PCIE1_PWREN_A		O	
		PCIE3_PWREN		O	
AG17	GPIO_13		P3	B	Configurable I/O
		GSBI4_0		B	GSBI4[0]
		PCIE2_PWRFLT_N_A		I	Power fault indication
		PCIE1_PWRFLT_N_A		I	
		PCIE3_PWRFLT_N		I	
B11	GPIO_14		P3	B	Configurable I/O
		AUDIO_PCM_DOUT		Z	Audio PCM data output
		BOOT_FROM_ROM		I	Select option to boot from ROM.
C11	GPIO_15		P3	B	Configurable I/O
		AUDIO_PCM_DIN		I	Audio PCM data input
B10	GPIO_16		P3	B	Configurable I/O
		AUDIO_PCM_SYNC		B	Audio PCM sync
A10	GPIO_17		P3	B	Configurable I/O
		AUDIO_PCM_CLK		B	Audio PCM clock
D11	GPIO_18		P3	B	Configurable I/O
		GSBI5_3		B	GSBI5[3]
C10	GPIO_19		P3	B	Configurable I/O
		GSBI5_2		B	GSBI5[2]
A9	GPIO_20		P3	B	Configurable I/O
		GSBI5_1		B	GSBI5[1]
D10	GPIO_21		P3	B	Configurable I/O
		GSBI5_0		B	GSBI5[0]
AH15	GPIO_22		P3	B	Configurable I/O
		GSBI2_3		B	GSBI2[3]
AH14	GPIO_23		P3	B	Configurable I/O
		GSBI2_2		B	GSBI2[2]

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
AH13	GPIO_24		P3	B	Configurable I/O
		GSBI2_1		B	GSBI2[1]
AH12	GPIO_25		P3	B	Configurable I/O
		GSBI2_0		B	GSBI2[0]
AG6	GPIO_26		P3	B	Configurable I/O
C23	GPIO_27		P3	B	Configurable I/O
		MI2S_WS_A		B	Word select
		RGMII2_CK_TX		Z	Transmit reference clock.
		GSBI6_A_3		B	GSBI6_A[3]
B25	GPIO_28		P3	B	Configurable I/O
		MI2S_SCLK_A		B	Bit clock
		RGMII2_TX_3		Z	Transmit data.
		GSBI6_A_2		B	GSBI6_A[2]
A25	GPIO_29		P3	B	Configurable I/O
		MI2S_MCLK_A		O	Master clock
		RGMII2_TX_2		Z	Transmit data.
		GSBI6_A_1		B	GSBI6_A[1]
B24	GPIO_30		P3	B	Configurable I/O
		MI2S_DATA_SD2		B	Serial data channel 2
		RGMII2_TX_1		Z	Transmit data.
		GSBI6_A_0		B	GSBI6_A[0]
A24	GPIO_31		P3	B	Configurable I/O
		MI2S_DATA_SD1		B	Serial data channel 1
		RGMII2_TX_0		Z	Transmit data.
C24	GPIO_32		P3	B	Configurable I/O
		MI2S_DATA_SD0_A		B	Serial data channel 0
		RGMII2_CTL_TX		Z	Transmit control.
D24	GPIO_33		P3	B	Configurable I/O
		MI2S_DATA_SD3		B	Serial data channel 3
		BOOT_CONFIG_4		I	BOOT_CONFIG[4]
AF9	GPIO_34		P3	B	Configurable I/O
		EBI2_NAND_CS_N		O	NAND chip select
		BOOT_CONFIG_3		I	BOOT_CONFIG[3]
AH6	GPIO_35		P3	B	Configurable I/O
		EBI2_CLE_N		O	
		BOOT_CONFIG_2		I	BOOT_CONFIG[2]

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
AG8	GPIO_36		P3	B	Configurable I/O
		EBI2_ALE_N		O	
		BOOT_CONFIG_1		I	BOOT_CONFIG[1]
AF8	GPIO_37		P3	B	Configurable I/O
		EBI2_WE_N		O	
		BOOT_CONFIG_6		I	BOOT_CONFIG[6]
AH9	GPIO_38		P3	B	Configurable I/O
		EBI2_OE_N		O	
		SDC1_DATA_7		B	Data
		BOOT_CONFIG_5		I	BOOT_CONFIG[5]
AH7	GPIO_39		P3	B	Configurable I/O
		EBI2_BUSY_N		I	
		SDC1_DATA_6		B	Data
AG7	GPIO_40		P3	B	Configurable I/O
		EBI2_A_D_7		B	Data
		SDC1_DATA_3		B	Data
AH8	GPIO_41		P3	B	Configurable I/O
		EBI2_A_D_6		B	Data
		SDC1_DATA_2		B	Data
AG9	GPIO_42		P3	B	Configurable I/O
		EBI2_A_D_5		B	Data
		SDC1_CLK		B	Clock
AE10	GPIO_43		P3	B	Configurable I/O1
		EBI2_A_D_4		B	Data
		SDC1_DATA_1		B	Data
AF10	GPIO_44		P3	B	Configurable I/O
		EBI2_A_D_3		B	Data
		SDC1_DATA_0		B	Data
AG10	GPIO_45		P3	B	Configurable I/O
		EBI2_A_D_2		B	Data
		SDC1_CMD		B	Command
AF11	GPIO_46		P3	B	Configurable I/O
		EBI2_A_D_1		B	Data
		SDC1_DATA_5		B	Data

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
AE11	GPIO_47		P3	B	Configurable I/O
		EBI2_A_D_0		B	Data
		SDC1_DATA_4		B	Data
AE27	GPIO_48		P3	B	Configurable I/O
		PCIE2_RST_N		O	Reset
		PCIE2_PRSNT_2_N_B		I	Power presence detection
		SPDIF_TX_A		O	SPDIF transmit
		BOOT_CONFIG_0		I	BOOT_CONFIG[0]
AE28	GPIO_49		P3	B	Configurable I/O
		PCIE2_WAKE_N		I	PCIe link reactivation
		PCIE2_PWREN_N_B		O	PCIe signal to power up/down the transceiver
		PCIE2_PWREN_B		O	
AF27	GPIO_50		P3	B	Configurable I/O
		PCIE2_CLK_REQ		I	Clock request from endpoint
		PCIE2_PWRFLT_N_B		I	Power fault indication
D23	GPIO_51		P3	B	Configurable I/O
		GSBI1_3		B	GSBI1[3]
		RGMII2_CK_RX		I	Receive reference clock.
D22	GPIO_52		P3	B	Configurable I/O
		GSBI1_2		B	GSBI1[2]
		RGMII2_RX_3		I	Receive data.
AG26	GPIO_53		P3	B	Configurable I/O
		GSBI1_1		B	GSBI1[1]
AF28	GPIO_54		P3	B	Configurable I/O
		GSBI1_0		B	GSBI1[0]
A26	GPIO_55		P3	B	Configurable I/O
		TSIF1_CLK		I	
		MI2S_WS_B		B	Word select
		GSBI6_B_3		B	GSBI6_B[3]
B26	GPIO_56		P3	B	Configurable I/O
		TSIF1_EN		I	
		MI2S_SCLK_B		B	Bit clock
		GSBI6_B_2		B	GSBI6_B[2]

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Table 2-39 GPIO Interface (cont.)

Pin #	Pin name	Configurable function	Voltage	Type	Functional description
C26	GPIO_57		P3	B	Configurable I/O
		TSIF1_DATA		I	
		MI2S_MCLK_B		O	Master clock
		GSBI6_B_1		B	GSBI6_B[1]
D26	GPIO_58		P3	B	Configurable I/O
		TSIF1_SYNC		I	
		MI2S_DATA_SD0_B		B	Serial data channel 0
		GSBI6_B_0		B	GSBI6_B[0]
C22	GPIO_59		P3	B	Configurable I/O
		TSIF2_CLK		I	
		RGMII2_RX_2		I	Receive data.
D21	GPIO_60		P3	B	Configurable I/O
		TSIF2_EN		I	
		RGMII2_RX_1		I	Receive data.
C21	GPIO_61		P3	B	Configurable I/O
		TSIF2_DATA		I	
		RGMII2_RX_0		I	Receive data.
		GSBI5_SPI_CS1B_N		Z	Slave select 1
C20	GPIO_62		P3	B	Configurable I/O
		TSIF2_SYNC		I	
		RGMII2_CTL_RX		I	Receive control.
		GSBI5_SPI_CS2B_N		Z	Slave select 2
AF22	GPIO_63		P3	B	Configurable I/O
		PCIE3_RST_N		O	
AG22	GPIO_64		P3	B	Configurable I/O
		PCIE3_WAKE_N		I	
AF21	GPIO_65		P3	B	Configurable I/O
		PCIE3_CLK_REQ		I	Clock request from endpoint
A11	GPIO_66		P3	B	Configurable I/O
AH23	GPIO_67		P4	B	Configurable I/O
		USB2_HSIC_STROBE		B	HSIC strobe
AG23	GPIO_68		P4	B	Configurable I/O
		USB2_HSIC_DATA		B	HSIC data

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3 Electrical Specifications

More specifications will be added in future revisions of this document.

3.1 Absolute maximum ratings

Operating the IPQ8064 under conditions beyond its absolute maximum ratings (listed in Table 3-1) may damage the device. Absolute maximum ratings are limiting values to be considered individually when all other parameters are within their specified operating ranges. Functional operation and specification compliance under any absolute maximum condition, or after exposure to any of these conditions, are not guaranteed or implied. Exposure may affect device reliability.

Table 3-1 Absolute maximum ratings

Parameter		Min	Max	Unit
	Junction temperature under bias	–	110	°C
	Case temperature under bias	–	110	°C
	Storage temperature	–	TBD	°C
Power-supply voltages				
VDDPX_1	Voltage with respect to VSSX_0	–	TBD	V
VDDPX_2	Voltage with respect to VSSX_0	–	TBD	V
VDDPX_3	Voltage with respect to VSSX_0	–	TBD	V
VDDPX_4	Voltage with respect to VSSX_0	–	TBD	V
VDDPX_5	Voltage with respect to VSSX_0	–	TBD	V
VDDPX_6	Voltage with respect to VSSX_0	–	TBD	V
Core voltages				
VDDCX_1		–	TBD	V
VDD_APC*		–	TBD	V
VDD_UBI32		–	TBD	V
Signal pins				
		–	TBD	V
		–	TBD	V
		–	TBD	mA

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3.2 Recommended operating conditions

Operating conditions include parameters that are under the control of the user: power-supply voltage and ambient temperature. The IPQ8064 meets all performance specifications when used within the recommended operating conditions (provided the absolute maximum ratings have never been exceeded).

Table 3-2 Recommended operating conditions

NOTE Tambient = 0°C min and Tcase = 110°C max.The maximum and minimum data are preliminary and are subject to change based on characterization results. Typical voltages are voltages at the BGA balls of the package. (For industrial temperature, contact your Qualcomm representative.)

Pin	Description	Min	Typ[1]	Max	Unit
Core voltages					
VDD_APC*	Applications processor cores	Refer to Table 3-3			
VDD_UBI32	Network accelerator power				
	turbo (733 MHz max)	1.10	1.15	1.20	V
	nominal (550 MHz max)	1.00	1.05	1.15	
VDDCX_1	Digital core				
	DDR freq = 533 MHz max	1.10	1.15	1.20	V
	DDR freq = 400 MHz max	1.05	1.10	1.15	V
Power-supply voltages					
VDDPX_1	P1 - EBI1 Pad power	1.283	1.35	1.45	V
VDDPX_2	P2 - Dual-voltage SD Card (SDC3)				
	2.95 V	2.7	2.95	3.0	V
	1.8 V	1.65	1.8	1.95	V
VDDPX_3	P3 - General pad power: MODE, JTAG, I2S, SDC1, GPIOs	1.65	1.8	1.95	V
VDDPX_4	P4 - HSIC				
	Used as GPIO	1.65	1.8	1.95	V
	Used as HSIC	1.14	1.2	1.3	V
VDDPX_5	P5 - Dual-voltage UIM1 pad power				
	2.95 V	2.7	2.95	3.0	V
	1.8 V	1.65	1.8	1.95	V
VDDPX_6	P6 - RGMII pad power	2.3	2.5	2.7	V
QSGMII, RGMII					
VDDPX_VBIAS_RGMII	Pad Bias RGMII	0.8	0.85	0.9	V

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Table 3-2 Recommended operating conditions (cont.)

NOTE Tambient = 0°C min and Tcase = 110°C max.The maximum and minimum data are preliminary and are subject to change based on characterization results. Typical voltages are voltages at the BGA balls of the package. (For industrial temperature, contact your Qualcomm representative.)

Pin	Description	Min	Typ[1]	Max	Unit
VDDA_QSGMII_PLL, VDDA_QSGMII_RX, VDDA_QSGMII_TX, VDDA_QSGMII_DRV, VDDA_SGMII	Transmit, receive, PLL, and clock driver	1.05	1.1	1.2	V
VDDA_QSGMII_H18	Analog high-voltage power for bandgap and bias generation	1.7	1.8	1.9	V
SATA					
VDDD_SATA, VDDD_SATA_VP	SATA core and transmit supply	1.05	1.1	1.2	V
VDDA_SATA_VPH	SATA high-voltage power	1.62	1.8	1.98	V
USB1					
VDDA_USB1_HS_1P8, VDDA_USB1_SS_1P8	USB 1.8 V supply	1.62	1.8	1.98	V
VDDA_USB1_HS_3P3	USB 3.3 V supply	2.97	3.3	3.63	V
VDDD_USB1_HS	USB High-Speed analog and digital supply	1.05	1.1	1.2	V
VDDD_USB1_SS, VDDD_USB1_SS_VPTX	USB Super Speed core and TX supply	1.05	1.1	1.2	V
USB3					
VDDA_USB3_HS_1P8, VDDA_USB3_SS_1P8	USB 1.8 V supply	1.62	1.8	1.98	V
VDDA_USB3_HS_3P3	USB 3.3 V supply	2.97	3.3	3.63	V
VDDD_USB3_HS	USB High-Speed analog and digital supply	1.05	1.1	1.2	V
VDDD_USB3_SS, VDDD_USB3_SS_VPTX	USB Super Speed core and TX supply	1.05	1.1	1.2	V
PCIe					
VDD_PCIE1_VP, VDD_PCIE1_VPTX0, VDD_PCIE2_VP, VDD_PCIE2_VPTX0, VDD_PCIE3_VP, VDD_PCIE3_VPTX0	PCIe core and transmit	1.05	1.1	1.155	V
VDD_VPH_PCIE1, VDD_VPH_PCIE2, VDD_VPH_PCIE3	PCIe high-voltage supply	1.674	1.8	1.98	V
Bias, Calibration, Reference, PLLs					
VDD_SR_PLL	Power for PLL circuits	1.05	1.1	1.2	V
VDD_SR2_PLL, VDD_HF_PLL*	Power for PLL circuits	1.7	1.8	1.9	V
VDD_QFPROM[2]	Power for programming Q-fuses	1.65	1.8	1.95	V
VDDA_CDC_EBI1_*, VDD_CDC_SDC*	Calibration Delay Circuits	1.05	1.1	1.2	V
VREF_EBI1_*	EBI1 reference voltage	0.735	0.75	0.765	V
VDDPX_VBIAS_SDC, VDDPX_VBIAS_UIM	Pad Vbias	1.2	1.25	1.3	V

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1. Typical voltages are voltages at the BGA balls of the package.
2. VDD_QFPROM should remain at 0V unless fuses are meant to be programmed.

Table 3-3 PVS for VDDC_APC*

NOTE The voltages in this table are recommended settings of PMIC output voltage in mV

Freq (MHz)	PVS = b000	PVS = b001	PVS = b010	PVS = b011
384	1000	925	875	800
600	1050	975	925	850
800	1100	1025	995	900
1000	1150	1075	1025	950
1200	1200	1125	1075	1000
1400	1250	1175	1125	1050

3.3 Power sequencing

Power is supplied by external voltage regulators, which should be configured for the power-up sequence shown in Table 3-4. Any other supplies that are not critical to this eartly power-up sequence, can be powered on by software after this sequence is completed.

Table 3-4 Power-on sequence

Step	Rails grouped by regulator	Voltage	Comments
1	VDD_SR_PLL	1.05	PLLs
	VDDA_CDC_EBI*		Calibration delay circuit
	VDDA_QSGMII_DRV VDDA_QSGMII_PLL VDDA_QSGMII_TX VDDA_QSGMII_RX VDDA_SGMII		QSGMII PHY rail
	VDD_VP_PCIE* VDD_VPTX0_PCIE* VDDA_SATA VDDA_SATA_VP VDDA_USB*_SS_VPTX		PCIE SATA USB PHY rails
	VDDA_USB*_HS VDDA_USB*_SS		USB3 DVDD
	VDD_UBI32		Network accelerator core
	VDDCX_1 VDD_CDC_SDC*		Core SDC CDC circuits
	VDD_APC0, VDD_APC1		Kraits
1a	VDDPX_VBIAS_RGMII	0.85	Pad bias
	VDDPX_VBIAS_SDC VDDPX_VBIAS_UIM	1.25	SDC, UIM, RGMII

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Table 3-4 Power-on sequence

Step	Rails grouped by regulator	Voltage	Comments
2	VDDPX_1	1.35	DDR3
	VDDPX_3, VDDPX_4, VDDA_USB*_HS_1P8, VDDA_USB*_SS_1P8, VDDA_SATA_VPH, VDD_SR2_PLL, VDD_HF_PLL* VDDA_QSGMII_H18	1.8	GPIOs, HSIC, USB3, SATA, PLLs, QSGMII bandgap and bias
	VDDPX_6	2.5	RGMII
	VDDPX_2, VDDPX_5	2.95	SDC2 / eMMC, UIM1
	HS_USB_3.3	3.3	HS_USB
2a	QFPROM_FUSE_PROG	1.8	QFPROM[1]

1. The QFPROM voltage, if enabled in the hardware power sequence, should be the last step in the sequence. The expectation is to have this voltage off at start-up, and enabled by software GPIO control if and when required.

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3.4 Digital logic characteristics

Specifications for the digital I/Os depend on the pad voltage being used. Logic specifications are listed in Table 3-5, Table 3-6, Table 3-7, and Table 3-8for V_{DD_PX} = 1.8 V (P3 for most pad circuits), V_{DD_PX} = 1.2 V (P1 for EBI1), V_{DD_PX} = 2.85 V (SD card interface), and V_{DD_PX} = 2.5 V (RGMII interface), respectively.

Table 3-5 Digital I/O characteristics for V_{DD_PX} = 1.8 V nominal

Parameter		Comments	Min	Max	Unit
V_{IH}	High-level input voltage	CMOS/Schmitt	0.65 * V_{DD_PX}	V_{DD_PX} + 0.3	V
V_{IL}	Low-level input voltage	CMOS/Schmitt	-0.3	0.35 * V_{DD_PX}	V
V_{SHYS}	Schmitt hysteresis voltage		100	–	mV
I_{IH}	Input high leakage current [1, 2]	No pulldown	–	1	µA
I_{IL}	Input low leakage current [1, 2]	No pullup	-1	–	µA
I_{IHPD}	Input high leakage current [1, 3]	With pulldown	5	30	µA
I_{ILPU}	Input low leakage current [2, 3]	With pullup	-30	-5	µA
V_{OH}	High-level output voltage [4]	CMOS, at pin-rated drive strength	V_{DD_PX} - 0.45	V_{DD_PX}	V
V_{OL}	Low-level output voltage [4]	CMOS, at pin-rated drive strength	0	0.45	V
I_{OZH}	Tri-state leakage current [1]	Logic high output, no pulldown	–	1	µA
I_{OZL}	Tri-state leakage current [2]	Logic low output, no pullup	-1	–	µA
I_{OZHPD}	Tri-state leakage current [1, 3]	Logic high output with pulldown	5	30	µA
I_{OZLPU}	Tri-state leakage current [2, 3]	Logic low output with pullup	-30	-5	µA
I_{OZHKP}	Tri-state leakage current [1, 3]	Logic high output with keeper	-15	-3	µA
I_{OZLKP}	Tri-state leakage current [2, 3]	Logic low output with keeper	3	15	µA
I_{ISL}	Sleep crystal input leakage		-0.15	0.15	µA
I_{IHVKP}	High-V tolerant input leakage	With keeper	-1	–	µA
C_{IN}	Input capacitance [5]		–	5	pF

1. Pin voltage = V_{DD_PX} max. For keeper pins, pin voltage = V_{DD_PX} max - 0.45 V.
2. Pin voltage = GND and supply = V_{DD_PX} max. For keeper pins, pin voltage = 0.45 V and supply = V_{DD_PX} max.
3. Refer to Table 2-1 for pullup, pulldown, and keeper details.
4. Refer to Table 2-1 for each output pin's drive strength (I_{OH} and I_{OL}); the drive strengths of many output pins are programmable and depend on the associated supply voltage.
5. Input capacitance is guaranteed by design, but is not 100% tested.

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Table 3-6 Digital I/O characteristics for V_{DD_PX} = 1.2 V nominal

Parameter		Comments	Min	Max	Unit
V_{IH}	High-level input voltage [1]	CMOS	V_{ref} + 0.13 V	–	V
V_{IL}	Low-level input voltage [1]	CMOS	–	V_{ref} - 0.13 V	V
I_{IH}	Input high leakage current [2]	No pulldown	–	2	µA
I_{IL}	Input low leakage current [3]	No pullup	-2	–	µA
I_{IHPD}	Input high leakage current [2, 4]	With pulldown	40	200	µA
I_{ILPU}	Input low leakage current [3, 4]	With pullup	-200	-40	µA
V_{OH}	High-level output voltage [5]	CMOS, at pin rated drive strength	0.9 * V_{DD_PX}	–	V
V_{OL}	Low-level output voltage [5]	CMOS, at pin rated drive strength	–	0.1 * V_{DD_PX}	V
I_{OZH}	Tri-state leakage current	Logic high output	–	1	µA
I_{OZL}	Tri-state leakage current	Logic low output	-1	–	µA
C_{IN}	Input capacitance [6]		1	2	pF
$C_{I/O}$	I/O capacitance [6]	I/O, DQS, DQ, or clock pins	1.25	2.5	pF

1. V_{ref} = V_{DD_PX}/2.
2. Pin voltage = V_{DD_PX} max.
3. Pin voltage = GND and supply = V_{DD_PX} max.
4. Refer to Table 2-1 for pullup, pulldown, and keeper details.
5. Refer to Table 2-1 for each output pin's drive strength (I_{OH} and I_{OL}); the drive strengths of many output pins are programmable and depend on the associated supply voltage.
6. Input and I/O capacitances are guaranteed by design, but are not 100% tested.

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Table 3-7 Digital I/O characteristics for V_{DD_PX} = 2.85 V nominal (SD card interface)

Parameter		Comments	Min	Max	Unit
V_{IH}	High-level input voltage	CMOS/Schmitt	0.65 * V_{DD_PX}	V_{DD_PX} + 0.3	V
V_{IL}	Low-level input voltage	CMOS/Schmitt	-0.3	0.35 * V_{DD_PX}	V
V_{SHYS}	Schmitt hysteresis voltage		100	–	mV
I_{IH}	Input high leakage current [1, 2]	No pulldown	–	10	µA
I_{IL}	Input low leakage current [1, 2]	No pullup	-10	–	µA
I_{IHPD}	Input high leakage current [1, 3]	With pulldown	10	60	µA
I_{ILPU}	Input low leakage current [2, 3]	With pullup	-60	-10	µA
I_{OZH}	Tri-state leakage current [1]	Logic high output, no pulldown	–	10	µA
I_{OZL}	Tri-state leakage current [2]	Logic low output, no pullup	-10	–	µA
I_{OZHPD}	Tri-state leakage current [1, 3]	Logic high output with pulldown	10	60	µA
I_{OZHPU}	Tri-state leakage current [2, 3]	Logic high output with pullup	-60	-10	µA
I_{OZHKP}	Tri-state leakage current [1, 3]	Logic high output with keeper	-25	-5	µA
I_{OZLKP}	Tri-state leakage current [2, 3]	Logic low output with keeper	5	25	µA
V_{OH}	High-level output voltage [4]	CMOS, at pin rated drive strength	V_{DD_PX} - 0.45	V_{DD_PX}	V
V_{OL}	Low-level output voltage [4]	CMOS, at pin rated drive strength	0	0.45	V
C_{IN}	Input capacitance [5]		–	5	pF

1. Pin voltage = V_{DD_PX} max. For keeper pins, pin voltage = V_{DD_PX} max - 0.45 V.
2. Pin voltage = GND and supply = V_{DD_PX} max. For keeper pins, pin voltage = 0.45 V and supply = V_{DD_PX} max.
3. Refer to Table 2-1 for pullup, pulldown, and keeper details.
4. Refer to Table 2-1 for each output pin's drive strength (I_{OH} and I_{OL}); the drive strengths of many output pins are programmable and depend on the associated supply voltage.
5. Input capacitance is guaranteed by design, but is not 100% tested.

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Table 3-8 Digital I/O characteristics for V_{DD_PX} = 2.5 V nominal (RGMII interface)

Parameter		Comments	Min	Max	Unit
V_{IH}	High-level input voltage	CMOS/Schmitt	1.7	–	V
V_{IL}	Low-level input voltage	CMOS/Schmitt	–	0.7	V
V_{SHYS}	Schmitt hysteresis voltage		200	400	mV
I_{IH}	Input high leakage current [1, 2]	No pulldown	–	15	µA
I_{IL}	Input low leakage current [1, 2]	No pullup	-15	–	µA
I_{IHPD}	Input high leakage current [1, 3]	With pulldown	10	60	µA
I_{ILPU}	Input low leakage current [2, 3]	With pullup	-60	-10	µA
I_{OZH}	Tri-state leakage current [1]	Logic high output, no pulldown	–	15	µA
I_{OZL}	Tri-state leakage current [2]	Logic low output, no pullup	-15	–	µA
I_{OZHPD}	Tri-state leakage current [1, 3]	Logic high output with pulldown	5	70	µA
I_{OZHPU}	Tri-state leakage current [2, 3]	Logic high output with pullup	-70	-5	µA
I_{OZHKP}	Tri-state leakage current [1, 3]	Logic high output with keeper	-70	-5	µA
I_{OZLKP}	Tri-state leakage current [2, 3]	Logic low output with keeper	5	70	µA
V_{OH}	High-level output voltage [4]	CMOS, at pin rated drive strength	2.0	V_{DD_PX}+0.3	V
V_{OL}	Low-level output voltage [4]	CMOS, at pin rated drive strength	-0.3	0.4	V
C_{IN}	Input capacitance [5]		–	5	pF

1. Pin voltage = V_{DD_PX} max. For keeper pins, pin voltage = V_{DD_PX} max - 0.45 V.
2. Pin voltage = GND and supply = V_{DD_PX} max. For keeper pins, pin voltage = 0.45 V and supply = V_{DD_PX} max.
3. Refer to Table 2-1 for pullup, pulldown, and keeper details.
4. Refer to Table 2-1 for each output pin's drive strength (I_{OH} and I_{OL}); the drive strengths of many output pins are programmable and depend on the associated supply voltage.
5. Input capacitance is guaranteed by design, but is not 100% tested.

In all digital I/O cases, V_{OL} and V_{OH} are linear functions (see Figure 3-1) with respect to the drive current (see Table 2-1). They can be calculated using these relationships:

$$Vol[\max] = \frac{\%drive \times 450}{100} mV$$

$$Voh[\min] = Vdd_px - \left(\frac{\%drive \times 450}{100}\right) mV$$

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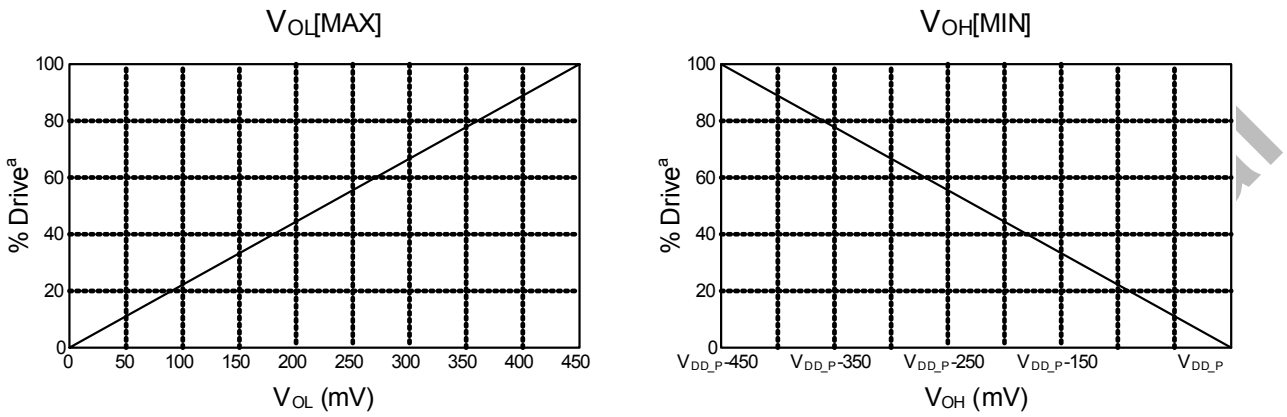


Figure 3-1 IV curve for V_{OL} and V_{OH} (valid for all V_{DD_PX})

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3.5 Timing characteristics

Specifications for the device timing characteristics are included (where appropriate) under each function's section, along with all its other performance specifications. Some general comments about timing characteristics are included here.

NOTE All IPQ8064 devices are characterized with actively terminated loads, so all baseband timing parameters in this document assume no bus loading. This is described further in Section 3.5.2.

3.5.1 Timing diagram conventions

The conventions used throughout this document for timing diagrams are shown in Figure 3-2. For each signal in the diagram:

- One clock period (T) extends from one rising clock edge to the next rising clock edge.
- The high level represents 1, the low level represents 0, and the middle level represents the floating (high-impedance) state.
- When both the high and low levels are shown, the meaning depends on the signal.
- A single signal indicates *don't care*.
- In the case of bus activity, if both high and low levels are shown, this indicates that the processor or external interface is driving a value, but that this value may or may not be valid.

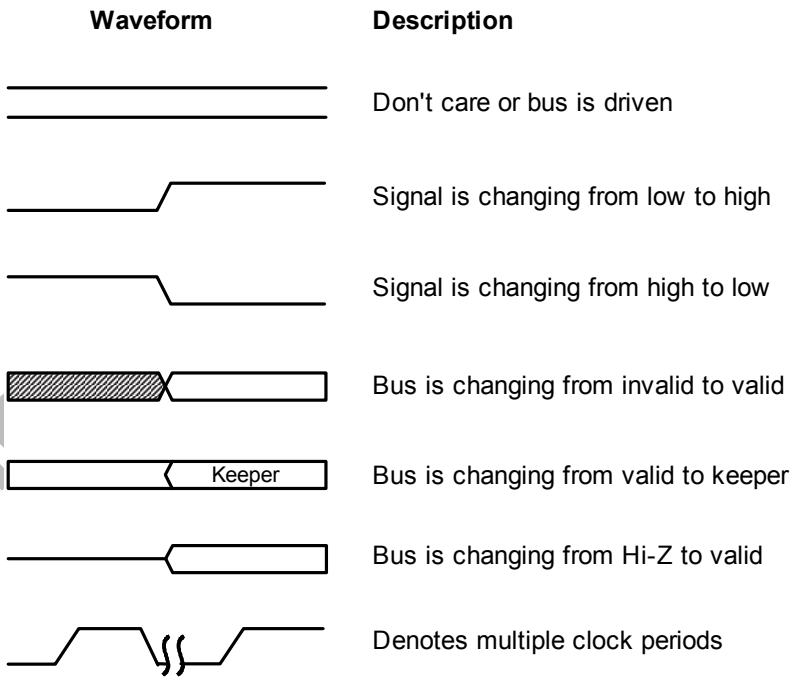


Figure 3-2 Timing diagram conventions

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3.5.2 Rise and fall time specifications

The testers that characterize IPQ8064 have actively terminated loads, making the rise and fall times quicker (mimicking a no-load condition). The impact that different external load conditions have on rise and fall times is shown in Figure 3-3.

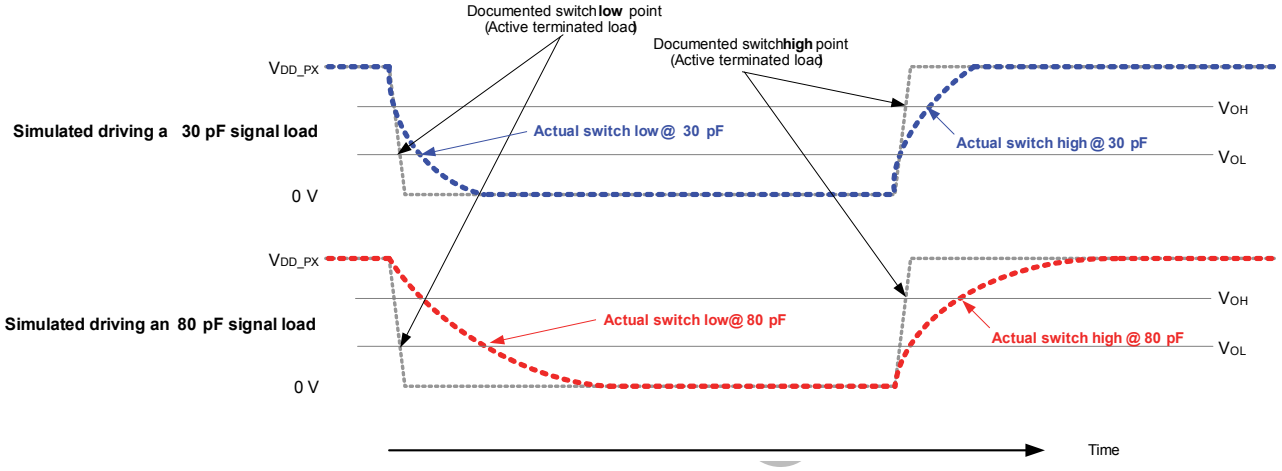


Figure 3-3 Rise and fall times under different load conditions

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3.6 Memory support

NOTE Information contained in this section is preliminary and is subject to change.

3.6.1 EBI1

All timing parameters in this document assume no bus loading. Rise/fall times must be factored into the numbers in this document. For example, setup times will get worse and hold times may get better.

3.6.1.1 EBI1 pad drive strengths

Pads for EBI1 are tailored for the 1.35 V interface and are source-terminated. Before the source termination, the pad drive strength is 15 mA to 60 mA. But at the pads, after the source termination, the drive strength at IOL, IOH is equivalent to 1.3 mA to 5.15 mA for PCDDR3 configuration in non-linear steps when the JEDEC standard range (90% to 10%) is followed.

3.6.1.1.1 PCDDR3 SDRAM clock

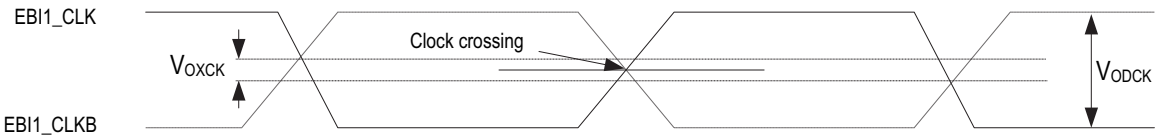


Figure 3-4 PCDDR3 SDRAM differential clock signal

Table 3-9 PCDDR3 SDRAM differential clock timing parameters

Parameter		Min	Max	Unit
1/t(ck)	PCDDR3 clock frequency	400	533	MHz
V_{IXCK}	Differential input cross-point voltage relative to VDD/2 for CK, CK#	-0.15	0.15	V
V_{ODCK}	AC differential output voltage	0.35		V

3.6.1.1.2 PCDDR3 SDRAM strobe signal

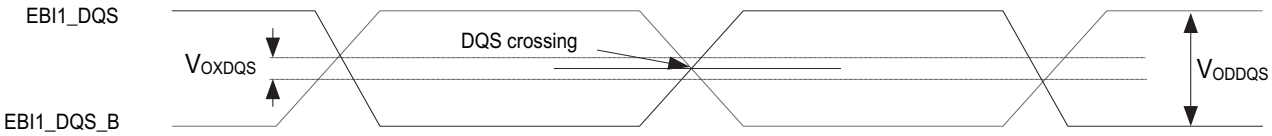


Figure 3-5 PCDDR3 SDRAM differential strobe signals

Table 3-10 PCDDR3 SDRAM differential strobe timing parameters

Parameter		Min	Max	Unit
V_{ODDQS}	AC differential output voltage	–	0.35	V
V_{IXDQS}	Differential input cross-point voltage relative to VDD/2 for DQS, DQS#	-0.15	0.15	V

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3.6.1.1.3 PCDDR3 SDRAM timing

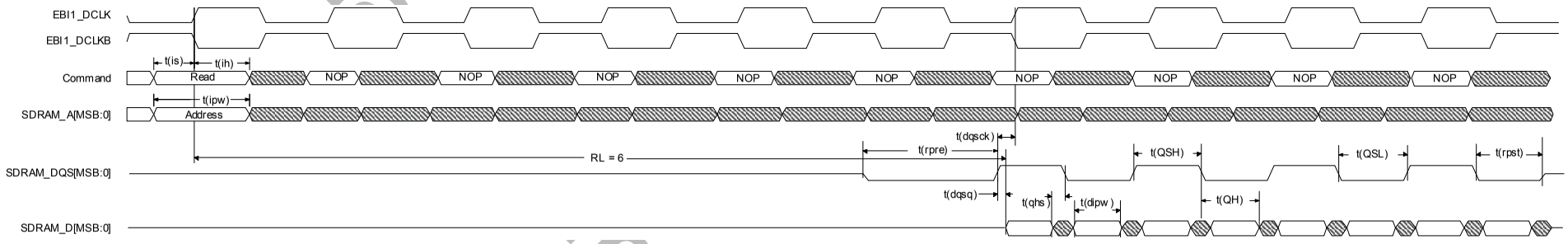


Figure 3-6 PCDDR3 SDRAM read timing

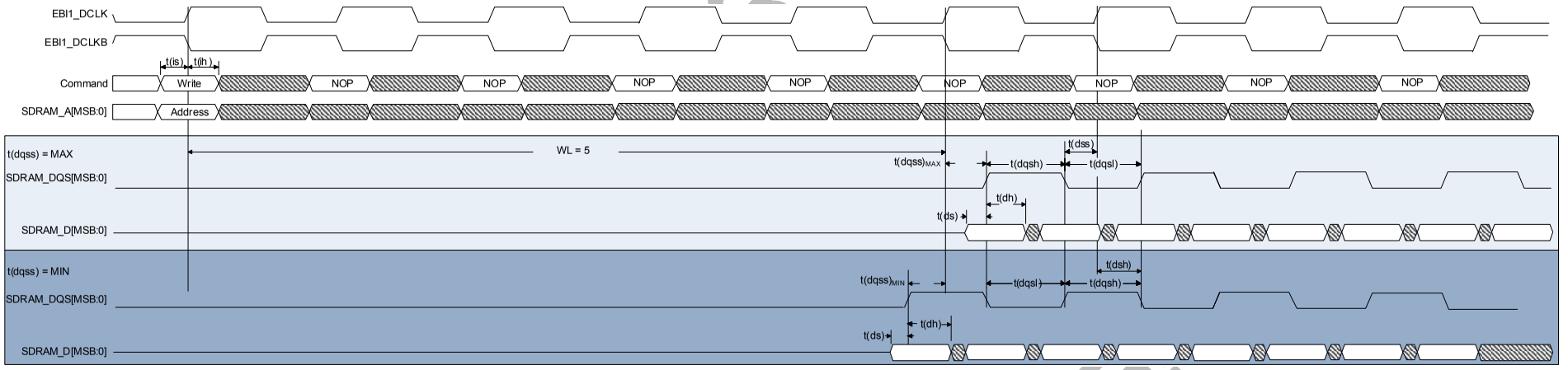


Figure 3-7 PCDDR3 SDRAM write timing

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Table 3-11 PCDDR3/PCDDR3L SDRAM memory specification

Parameter	Symbol	Min	Max	Units
Clock high pulse width	tCH	0.43	–	tCK(avg)
Clock low pulse width	tCL	0.43	–	tCK(avg)
Data timing				
DQS, DQS# to DQ skew, per group, per access	tDQSQ	-	200	ps
DQ output hold time from DQS, DQS#	tQH	0.38	–	tCK(avg)
Data setup time to DQS, DQS# referenced to Vih(ac)/Vil(ac) levels	tDS			
PCDDR3		175	–	ps
PCDDR3L		190	–	ps
Data hold time from DQS, DQS# referenced to Vih(dc)/Vil(dc) levels	tDH			
PCDDR3		200	–	ps
PCDDR3L		210	–	ps
DQ and DM input pulse width for each input	tDIPW	490	–	ps
Data strobe timing				
DQS, DQS# differential input low pulse width	tDQSL	0.45	0.55	tCK(avg)
DQS, DQS# differential input high pulse width	tDQSH	0.45	0.55	tCK(avg)
DQS, DQS# rising edge to CK, CK# rising edge	tDQSS	-0.25	0.25	tCK(avg)
DQS, DQS# falling-edge setup time to CK, CK# rising edge	tDSS	0.2	–	tCK(avg)
DQS, DQS# falling-edge hold time from CK, CK# rising edge	tDSH	0.2	–	tCK(avg)
Command and address timing				
Command and address setup time to CK, CK# referenced to Vih(ac)/Vil(ac) levels	tIS			
PCDDR3		375	–	ps
PCDDR3L		390	–	ps
Command and address hold time from CK, CK# referenced to Vih(dc)/Vil(dc) levels	tIH			
PCDDR3		300	–	ps
PCDDR3L		310	–	ps
Control and address input pulse width for each input	tIPW	780	–	ps

3.6.2 EBI2 NAND interface

Each access to the NAND flash device involving the controller executes a sequence of signal assertion and de-assertion. The state-machine registers are designed to allow up to seven distinct configurations, including wait states.

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The register settings dictate the signal status once the state-machine is started. For example, CMD_CLE_EN (FLASH_XFR_STEPx[22]) values in the step controls the CLE pin state (1: high, 0: low). If the controller must send a multiple command sequence, it may not need to do all the steps. It can start from the beginning of the sequence and, after sending the command, it can loop back and send the next command without finishing the sequence.

FLASH_XFR_STEPx[31:30] – CMD_SEQ_STEP_NUMBER and
FLASH_XFR_STEPx[15:14] – DATA_SEQ_STEP_NUMBER defines the step number in the command/data sequence:

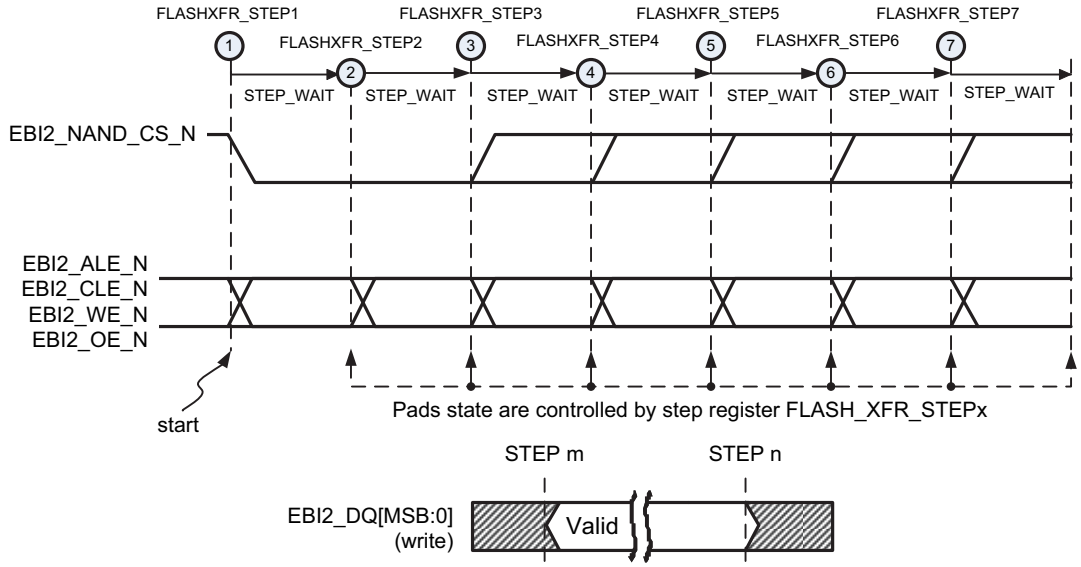
00 => Simple step
01 => Loop start
10 => Loop end
11 => Last step

If a value of "11" is programmed to the SEQ_STEP_NUMBER register, it indicates the step is the last step in the sequence and the controller will end the sequence after finishing the step. The timing parameters in Table 3-12 use a four-step configuration as an example. If the step configuration is changed, the timing parameters that mapped to the registers should be adjusted accordingly. In step configuration registers (FLASH_XFR_STEPx), the upper 16 bits [31:16] configure the NAND *command* and *address* cycles, whereas the lower 16 bits [15:0] configure the *data* read and write cycles.

Table 3-12 NAND timing parameters

Parameter	Description
T	EBI2 CLK cycle; timing parameters are based upon a 4-step configuration, FLASH_XFR_STEP1 to FLASH_XFR_STEP4.
a	FLASH_XFR_STEP1[29:26] + 1
b	FLASH_XFR_STEP2[29:26] + 1
c	FLASH_XFR_STEP3[29:26] + 1
d	FLASH_XFR_STEP4[29:26] + 1
e	FLASH_XFR_STEP1[13:10] + 1
f	FLASH_XFR_STEP2[13:10] + 1
g	FLASH_XFR_STEP3[13:10] + 1
h	FLASH_XFR_STEP4[13:10] + 1
i	NAND_DEVn_CFG1[4:2]+1
k	FLASH_XFR_STEP2[13:10] + FLASH_XFR_STEP2[1:0] + 1
m	FLASH_XFR_STEP3[13:10] + FLASH_XFR_STEP3[1:0] + 1
n	(NAND_DEVn_CFG[22:17] + 1) × 2

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FLASH_XFR_STEPx (x = 1 to 7)			* Not used
Command/Address		**Data**	
31:30	CMD_SEQ_SETP_NUM	15:14	DATA_SEQ_STEP_NUMBER
29:26	CMD_STEP1_WAIT	13:10	DATA_STEP1_WAIT
25	CMD_AOUT_EN*	9	DATA_AOUT_EN*
24	CMD_DATA_EN	8	DATA_DATA_EN
23	CMD_CE_EN	7	DATA_CE_EN
22	CMD_CLE_EN	6	DATA_CLE_EN
21	CMD_ALE_PIN	5	DATA_ALE_PIN
20	CMD_WE_EN	4	DATA_WE_EN
19	CMD_RE_EN	3	DATA_RE_EN
18	CMD_WIDE	2	DATA_WIDE
17:16	RESERVED*	1:0	EXTA_READ_WAIT

Figure 3-8 NAND state machine registers

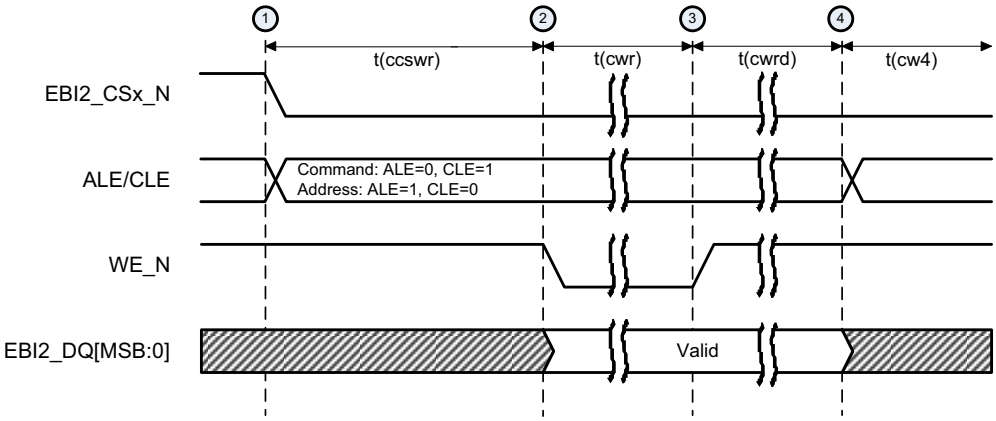


Figure 3-9 NAND command and address cycles

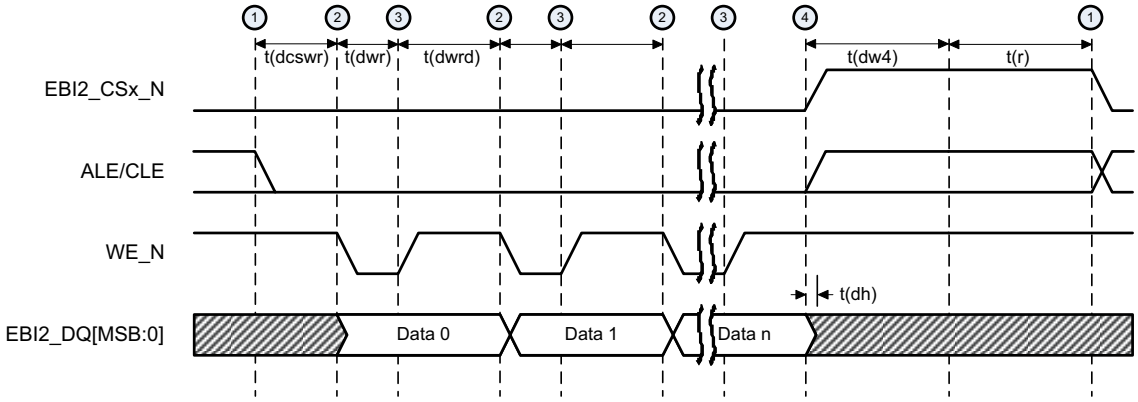
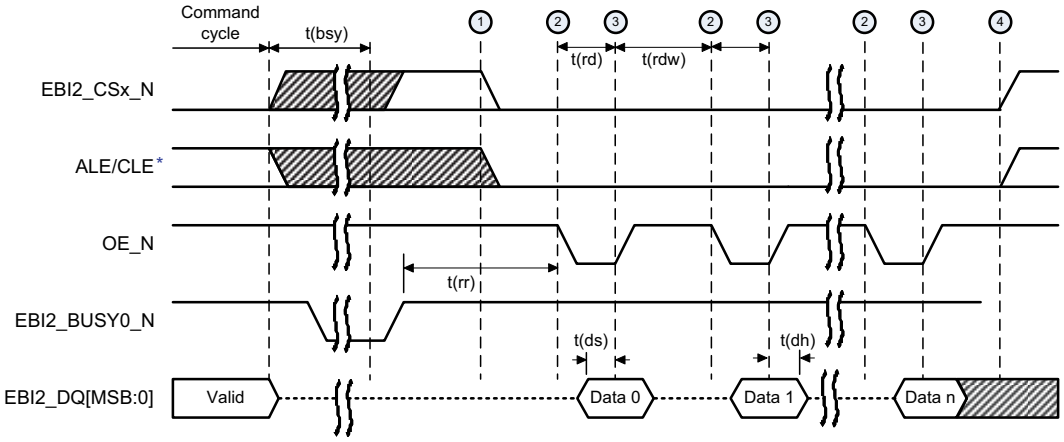


Figure 3-10 NAND data write



* During data read wait period, ALE and CLE stay low only if EBI2_CSx_N is driven low as required by the FLASH device.

Figure 3-11 NAND data read

Table 3-13 NAND timing – AC characteristics [1]

Parameter		Comments	Min	Max	Unit
Command/address cycles					
t(ccswr)	Chip-select active to write active		aT-5	–	ns
t(cwr)	WE_N active time		bT-5	–	ns
t(cwrd)	Data hold time		cT-5	–	ns
t(cw4)	Wait cycle		dT-5	–	ns
Data cycles					
t(dcswr)	Chip-select active to write active		eT-5	–	ns
t(dwr)	WE_N active time		fT-5	–	ns
t(dwrd)	Write-data hold time		gT-5	–	ns
t(dw4)	Wait cycle		hT-5	–	ns
t(r)	Recovery cycles		iT-5	–	ns
t(rr)	Busy to read delay		3T	–	ns
t(rd)	OE_N active time		kT-5	–	ns
t(rdw)	Read wait cycle		mT-5	–	ns
t(bsy)	Busy detect delay		nT-5	–	ns
t(ds)	Read-data setup time		15	–	ns
t(dh)	Read-data hold time		0	–	ns

1. Variables ‘a’ through ‘n’ are defined in Table 3-12.

3.6.3 eMMC NAND flash on SDIO

NAND flash can be supported via the SDIO ports. See Section 3.7.7 for SDIO details.

3.7 Connectivity

The connectivity functions supported by the IPQ8064 include:

- Two USB 3.0 ports
- Serial ATA (SATA) version 3.0 port
- Three PCI Express (PCIe) 2.0 ports
- Universal asynchronous receiver transmitter (UART) serial ports
- User identity module (UIM) ports
- Secure digital card controller (SDCC) ports
- Inter-integrated circuit (I^2C) interfaces for peripheral devices
- Multichannel I^2S (MI^2S) interface for digital audio support
- Serial peripheral interface (SPI) ports

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- Pulse code modulation (PCM) port
- Transport stream interface (TSIF) ports
- Media-independent interfaces QSGMII, SGMII, RGMII

Pertinent specifications for these functions — where appropriate — are stated in the following subsections.

NOTE The UART, UIM, I^2C, and SPI interfaces are multiplexed to output pins by properly configuring the six generic serial bus interface (GSBI).

In addition to the following hardware specifications, consult the latest software release notes for software-based performance features or limitations.

3.7.1 USB interfaces

Table 3-14 Summary of USB support

Applicable standard	Feature exceptions	IPQ8064 variations
Universal Serial Bus Specification, Revision 3.0	None	None
Universal Serial Bus Specification, Revision 2.0	None	For operating voltages, system clock, and VBUS – see Section 2.3.14 and Table 3-2

3.7.2 HSIC interface

Table 3-15 Supported HSIC standards and exceptions

Applicable standard	Feature exceptions	IPQ8064 variations
High-Speed Inter-Chip USB Electrical Specification, Version 1.0 (a supplement to the USB 2.0 specification – see Section 3.7.1)	Device mode not supported	None

3.7.3 SATA3 interface

Table 3-16 Summary of SATA3 support

Applicable standard	Feature exceptions	IPQ8064 variations
SATA Specification, Revision 3.0	None	None

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3.7.4 PCIe interface

Table 3-17 Summary of PCIe support

Applicable standard	Feature exceptions	IPQ8064 variations
PCI Express Specification, Revision 2.0 (January 15, 2007 or later)	None	None

Table 3-18 PCIE REFCLK DC specification and AC timing requirements

Symbol	Parameter	Min	Typ	Max	Unit
Period	Clock period	–	10	–	ns
DutyCycle	Clock duty cycle	40	–	60	%
V_diff	Output differential voltage (into the load Rload=100Ω)	300	–	–	mVppd
T_CC	Cycle to cycle jitter	–	–	150	ps
Z_{C-DC}	Clock source DC impedance	40	50	60	Ω
$V_{_CM}$	Clock source common mode voltage	250	–	550	mV

Table 3-19 PCIE REFCLK Jitter Specifications (for Common Refclk RX Architecture)

Symbol	Parameter	Min	Typ	Max	Unit
$T_{REFCLK_HF_RMS}$	1.5MHz - 50MHz RMS jitter after applying a step HPF function[1]	–	–	3.1	ps
$T_{REFCLK_LF_RMS}$	10KHz - 1.5MHz RMS jitter after applying a step BPF function[1]	–	–	3	ps

1. Refer to PCIE Express 2.0 Base Specification Chapter 4.3.7, page 311

Table 3-20 PCIE REFCLK Jitter Specifications (for Data Clocked RX Architecture)

Symbol	Parameter	Min	Typ	Max	Unit
$T_{REFCLK_HF_RMS}$	1.5 MHz - 50 MHz RMS jitter after applying a step HPF function[1]	–	–	4	ps
$T_{REFCLK_LF_RMS}$	10 KHz - 1.5 MHz RMS jitter after applying a step BPF function[1]	–	–	7.5	ps

1. Refer to PCIE Express 2.0 Base Specification Chapter 4.3.7, page 311

3.7.5 High-speed UART interface

Table 3-21 Summary of UART support

Applicable standard	Feature exceptions	IPQ8064 variations
EIA RS232-C	None	None

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3.7.6 UIM interface

Table 3-22 Summary of UIM support

Applicable standard	Feature exceptions	IPQ8064 variations
ISO/IEC 7816-3	None	None

3.7.7 Secure digital card controller interfaces

Table 3-23 Summary of SDCC support

Applicable standard	Feature exceptions	IPQ8064 variations
MultiMediaCard Host Specification version 4.4.1 and 4.5	None	Timing specifications – see Figure 3-12 and Figure 3-13
Secure Digital: Physical Layer Specification version 3.0	None	
SDIO Card Specification version 3.0	None	

Single Data Rate - SDR

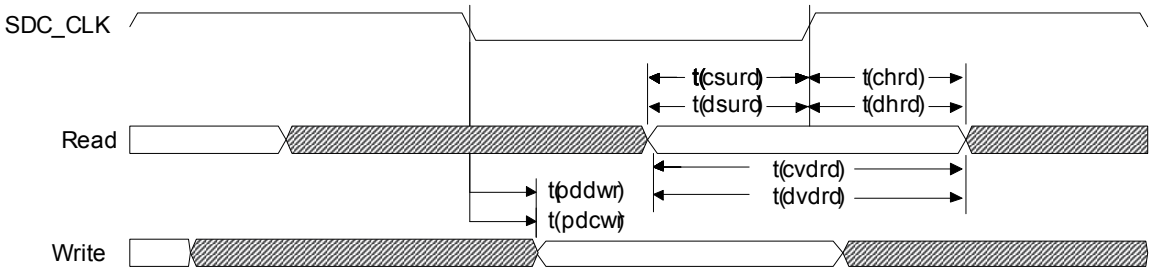


Figure 3-12 SDCC SDR timing waveforms

Double Data Rate - DDR

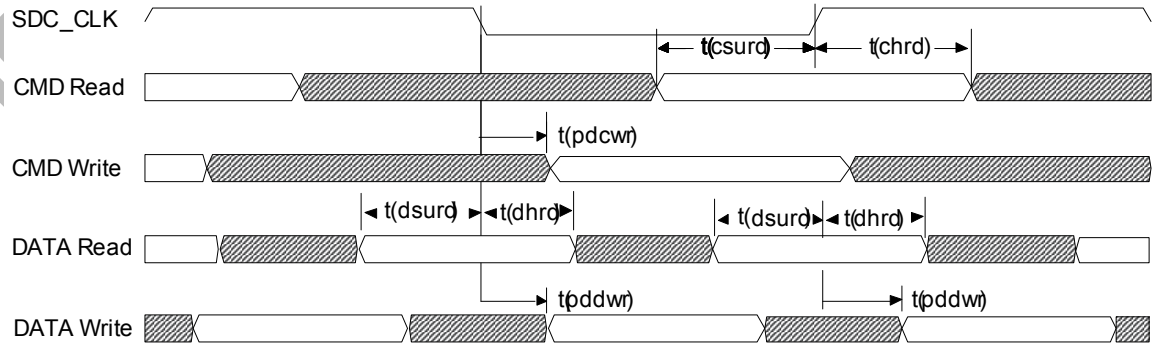


Figure 3-13 SDCC DDR timing waveforms

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Table 3-24 SDC1/SDC3 DDR mode timing parameters [1]

Parameter		Min	Max	Unit
t(chrd)	Command hold	1.5	–	ns
t(csurd)	Command setup	6.3	–	ns
t(dhrd)	Data hold	1.5	–	ns
t(dsurd)	Data setup	2	–	ns
t(pddwr)	Propagation delay on data write	0.8	6	ns
t(pdcwr)	Propagation delay on command write	2.5	3	ns

1. SDC1 and SDC3 DDR mode timing parameters are for clock frequencies up to 52 MHz.

Table 3-25 SDC1 SDR mode timing parameters [1]

Parameter		Min	Max	Unit
t(chrd)	Command hold	1.50	–	ns
t(csurd)	Command setup	2.50	–	ns
t(dhrd)	Data hold	1.50	–	ns
t(dsurd)	Data setup	2.50	–	ns
t(pddwr)	Propagation delay on data write	-3.7	1.50	ns
t(pdcwr)	Propagation delay on command write	-3.7	1.50	ns

1. SDC1 SDR timing parameters is for the clock frequencies up to 104 MHz.

Table 3-26 SDC1/SDC3 SDR104 mode timing parameters [1]

Parameter		Min	Max	Unit
t(cvdrd)	Command valid	2.50	–	ns
t(dvdrd)	Data valid	2.50	–	ns
t(pddwr)	Propagation delay on data write	-1.45	0.85	ns
t(pdcwr)	Propagation delay on command write	-1.45	0.85	ns

1. SDC1/SDC3 SDR104 mode timing parameters are for the clock frequencies up to 104 MHz.

Table 3-27 SDC2/SDC4 SDR33 mode timing parameters [1]

Parameter		Min	Max	Unit
t(chrd)	Command hold	1.50	–	ns
t(csurd)	Command setup	7.65	–	ns
t(dhrd)	Data hold	1.50	–	ns
t(dsurd)	Data setup	7.65	–	ns
t(pddwr)	Propagation delay on data write	-6.0	3.82	ns
t(pdcwr)	Propagation delay on command write	-6.0	3.82	ns

1. SDC2/SDC4 SDR33 mode timing parameters are for the clock frequencies up to 66 MHz.

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3.7.8 I2C interface

Table 3-28 Summary of I2C support

Applicable standard	Feature exceptions	IPQ8064 variations
II2C Specification, version 2.1, January 2000 (Philips Semiconductor document number 9398 393 40011)	■ High-speed mode (3.4 Mbps) is not supported. ■ 10-bit addressing is not supported. ■ Fast mode plus (1 Mbps) is not supported.	None

3.7.9 I2S interface

The IPQ8064 supports Multichannel I2S (MI2S) . The I2S interfaces meet the timing given in this section.

Table 3-29 Supported I2S standards and exceptions

Applicable standard	Feature exceptions	IPQ8064 variations
Philips Semiconductor, *I2S Bus Specification*, revised June 5, 1996		The IPQ8064 meets or exceeds this standard. The only exception is the IPQ8064 requires a 45/55 duty cycle when the SCK clock source is from an external source.

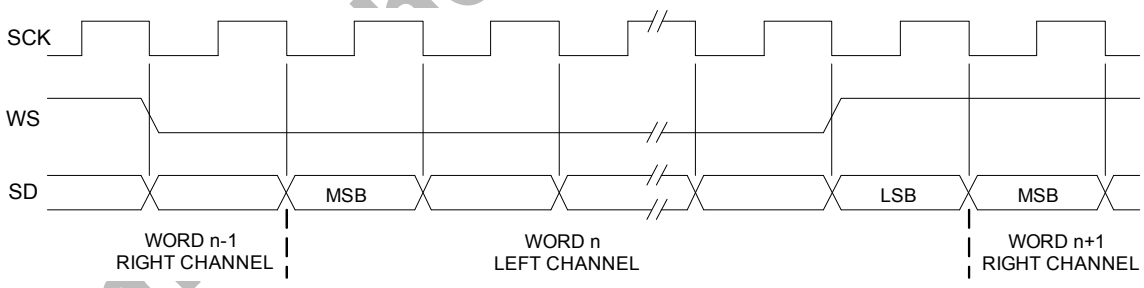


Figure 3-14 I2S interface basic timing

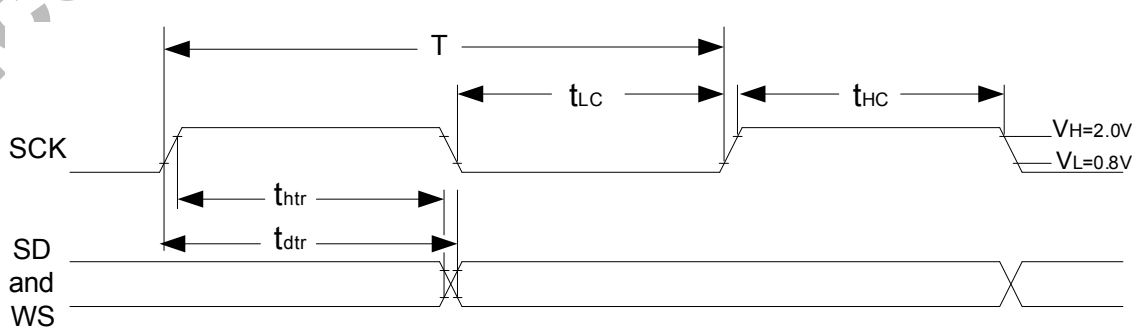


Figure 3-15 I2S interface transmitter timing

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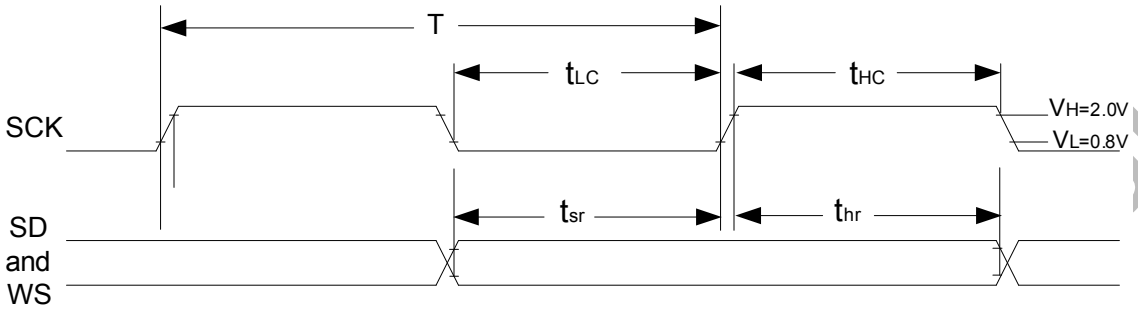


Figure 3-16 I²S interface receiver timing

Table 3-30 I²S interface timing using internal SCK clock

Parameter	Symbol	Min	Typ	Max	Unit	Remark
Clock SCK						
Frequency	f	0		12.288	MHz	C_L = 10 pF - 40 pF
Clock period	T	0		81.38	ns	C_L = 10 pF - 40 pF
Clock high	t_{HC}	0.45T		0.55T	ns	C_L = 10 pF - 40 pF
Clock low	t_{LC}	0.45T		0.55T	ns	C_L = 10 pF - 40 pF
Inputs SD*, WS						
Setup time	t_{sr}	16.276	–	–	ns	
Hold time	t_{hr}	0	–	–	ns	
Outputs SD*, WS						
Delay	t_{dtr}	–	–	65.1	ns	C_L = 10 pF - 40 pF
Hold time	t_{htr}	0	–	–	ns	C_L = 10 pF - 40 pF

Table 3-31 I²S interface timing using external SCK clock

Parameter	Symbol	Min	Typ	Max	Unit	Remark
Clock SCK						
Frequency	f	0		12.288	MHz	C_L = 10 pF - 40 pF
Clock period	T	0		81.38	ns	C_L = 10 pF - 40 pF
Clock high	t_{HC}	0.45T		0.55T	ns	C_L = 10 pF - 40 pF
Clock low	t_{LC}	0.45T		0.55T	ns	C_L = 10 pF - 40 pF
Inputs SD*, WS						
Setup time	t_{sr}	16.27	–	–	ns	
Hold time	t_{hr}	0	–	–	ns	
Outputs SD*, WS						
Delay	t_{dtr}	–	–	65.1	ns	C_L = 10 pF - 40 pF
Hold time	t_{htr}	0	–	–	ns	C_L = 10 pF - 40 pF

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3.7.10 Serial peripheral interface (master only)

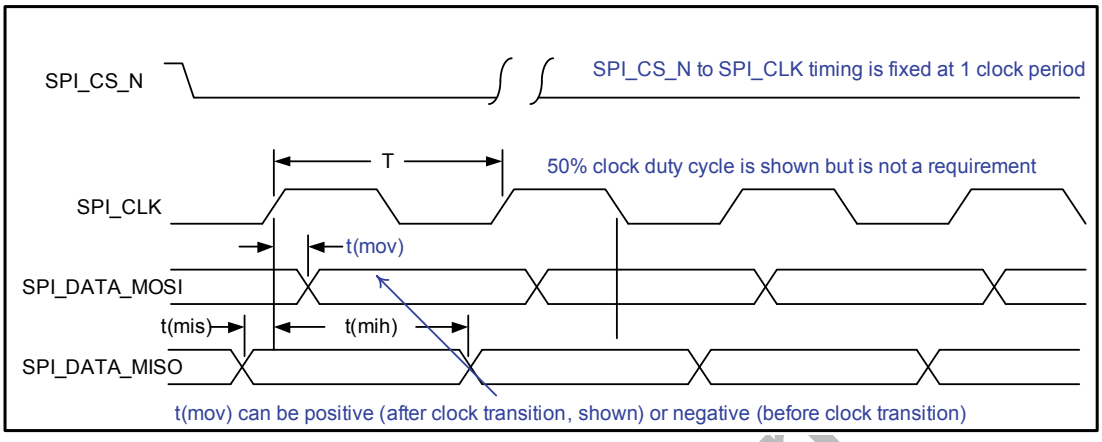


Figure 3-17 SPI master timing diagram

Table 3-32 SPI master timing characteristics (52 MHz max)

Parameter	Comments	Min	Typ	Max	Unit
SPI clock frequency		–	–	52	MHz
T (SPI clock period)		19.0	–	–	ns
t(ch)	Clock high	8.65	–	–	ns
t(cl)	Clock low	8.65	–	–	ns
t(mov)	Master output valid	-5	–	5	ns
t(mis)	Master input setup	5	–	–	ns
t(mih)	Master input hold	1	–	–	ns

3.7.11 QSGMII Interface

Table 3-33 QSGMII transmit DC electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit
T_BAUD	Transmit baud rate	–	5.000	–	Gbps
T_VDIFF[1]	Output differential voltage (into the load Rload=100Ω)	300	600	900	mVppd
T_Rs	Single-ended termination resistance	40	50	60	Ω
T_Vcm[2]	Output common mode voltage	0.6	0.8	1.05	V

1. The default value is 600 mVppd, and can be adjusted from 300 to 900 mVppd by register configuration.
2. The output common mode voltage changes accordingly if T_VDIFF is adjusted. It is also affected by supply voltage which is assumed 1.1 V+100mV/-50mV.

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Table 3-34 QSGMII receive DC electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit
R_BAUD	Receive baud rate	–	5.000	–	Gbps
R_VDIFF	Input differential voltage	100	–	1600[2]	mVppd
R_Rs	Single-ended termination resistance	40	50	60	Ω
R_Vcm[1]	Input common mode voltage	1.05	1.1	1.2	V

1. The input common mode voltage is affected by supply voltage which is assumed 1.1 V+100mV/-50mV.
2. The maximum differential voltage the receiver can tolerate.

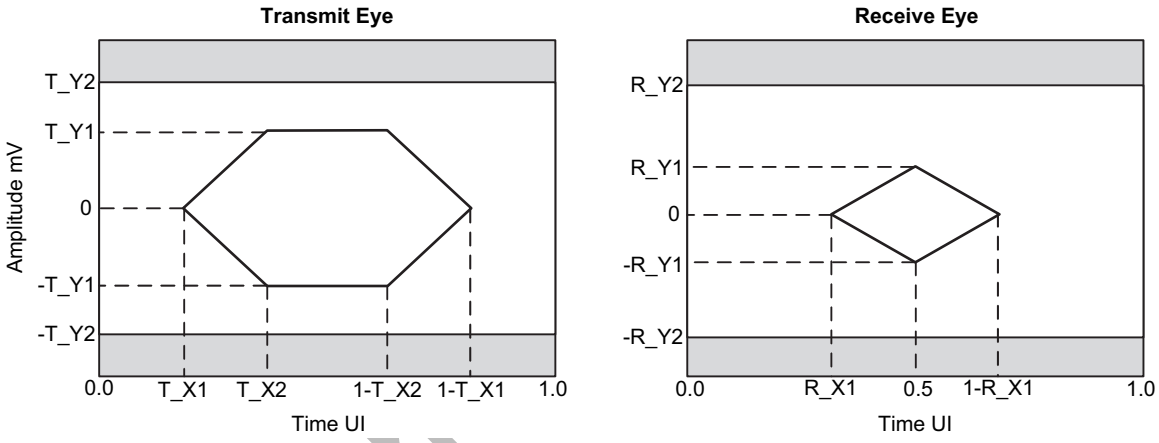


Figure 3-18 QSGMII jitter eye diagrams

Table 3-35 QSGMII transmit jitter specifications

Symbol	Parameter	Min	Typ	Max	Unit
T_UHPJ[1]	Uncorrelated high-probability jitter	–	–	0.15	UIpp
T_DCD	Duty cycle distortion	–	–	0.05	UIpp
T_Tj	Total jitter	–	–	0.30	UIpp
T_X1	Eye mask	–	–	0.15	UI
T_X2	Eye mask	–	–	0.40	UI
T_Y1	Eye mask	200	–	-	mV
T_Y2	Eye mask	–	–	450	mV

1. DJ component; no correlation to any signal level being transmitted.

Table 3-36 QSGMII receive jitter specifications

Symbol	Parameter	Min	Typ	Max	Unit
R_BHPJ[1]	Bounded high-probability jitter	–	–	0.45	UIpp
R_SJ_max	Sinusoidal jitter, maximum	–	–	5	UIpp

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Table 3-36 QSGMII receive jitter specifications (cont.)

Symbol	Parameter	Min	Typ	Max	Unit
R_SJ_hf	Sinusoidal jitter, high frequency	–	–	0.05	UIpp
R_TJ	Total jitter (sinusoidal jitter not included)	–	–	0.60	UIpp
R_X1	Eye mask	–	–	0.30	UI
R_Y1	Eye mask	50	–	-	mV
R_Y2	Eye mask	–	–	450	mV

1. The sum of uncorrelated, bounded, high-probability jitter (UBHPJ, 0.15UI) and correlated, bounded, high-probability jitter (CBHPJ, 0.3UI). UBHPJ shows no correlation to any signal level being transmitted. CBHPJ shows a strong correlation to the signal level being transmitted, so can be considered equalizable.

3.7.12 SGMII Interface

Table 3-37 SGMII transmit DC electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit
T_BAUD	Transmit baud rate	–	1.250	–	Gbps
T_V_{DIFF}[1]	Output differential voltage (into the load Rload=100Ω)	300	600	900	mVppd
T_R_s	Single-ended termination resistance	40	50	60	Ω
T_V_{cm}[2]	Output common mode voltage	0.6	0.8	1.05	V

1. The default value is 600 mVppd, and can be adjusted from 300 to 900 mVppd by register configuration.
2. The output common mode voltage changes accordingly if T_VDIFF is adjusted. It is also affected by supply voltage which is assumed 1.1 V+100mV/-50mV.

Table 3-38 SGMII receive DC electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit
R_BAUD	Receive baud rate	–	1.250	–	Gbps
R_V_{DIFF}	Input differential voltage	100	–	1600[2]	mVppd
R_R_s	Single-ended termination resistance	40	50	60	Ω
R_V_{cm}[1]	Input common mode voltage	1.05	1.1	1.2	V

1. The input common mode voltage is affected by supply voltage which is assumed 1.1 V+100mV/-50mV.
2. The maximum differential voltage the receiver can tolerate.

Table 3-39 SGMII transmit jitter specifications

Symbol	Parameter	Min	Typ	Max	Unit
T_DCD	Duty cycle distortion	–	–	30	ps
T_DDJ	Data dependent jitter	–	–	70	ps
T_T_j	Total jitter	–	–	0.375	UIpp

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Table 3-40 SGMII receive jitter specifications

Symbol	Parameter	Min	Typ	Max	Unit
R_T_j	Total jitter	–	–	0.625	UIpp

3.7.13 RGMII Interface

Multiplexing of data and control information takes advantage of both edges of the reference clocks, sending the lower 4 bits on the rising edge and the upper 4 bits on the falling edge. Control signals can be multiplexed into a single clock cycle using the same technique. (See Figure 3-19.)

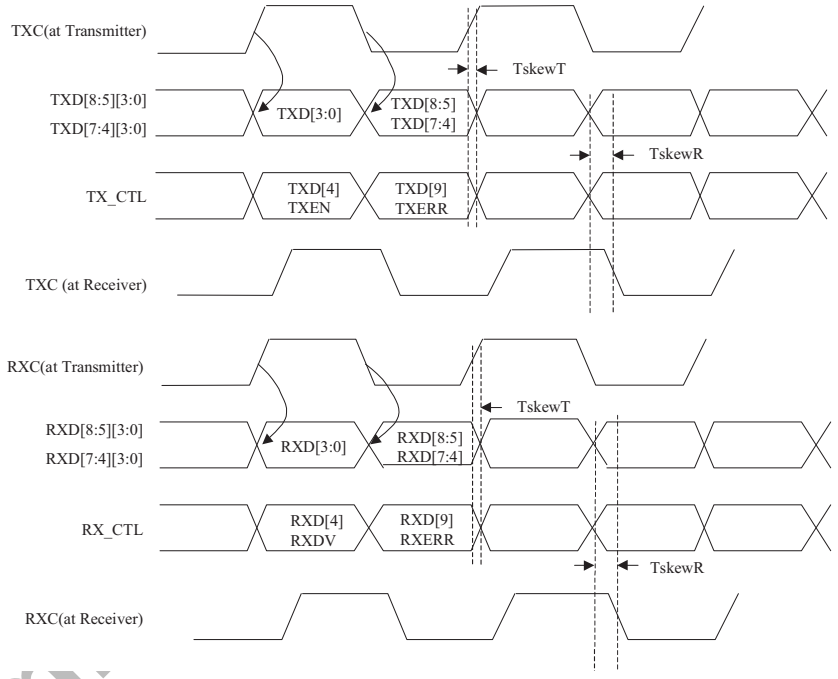


Figure 3-19 RGMII Multiplexing and Timing Diagram

Timing for this interface is such that the clock and data are generated simultaneously by the source of the signals; therefore, the skew between clock and data is critical to proper operation. This approach provides tighter control of skew. Timing values in Table 3-41 are defined in percentages of the nominal clock period so that the table is speed independent.

Table 3-41 RGMII speed-independent timing parameters

Symbol	Parameter	Min	Typical	Max	Units
TskewT	Data to Clock output Skew (at Transmitter)	-500	0	500	ps
TskewR	Data to Clock input Skew (at Receiver)[1]	1		2.6	ns
Tcyc	Clock Cycle Duration[2]	7.2	8	8.8	ns
Duty_G	Duty Cycle for Gigabit[3]	45	50	55	%

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Table 3-41 RGMII speed-independent timing parameters (cont.)

Symbol	Parameter	Min	Typical	Max	Units
Duty_T	Duty Cycle for 10/100T[3]	40	50	60	%
Tr / Tf	Rise / Fall Time (20-80%)			.75	ns

1. This implies clocks on the PC board are routed such that an additional trace delay of greater than 1.5 ns and less than 2.0 ns is be added to the associated clock signal. For 10/100 the Max value is unspecified.
2. For 10 Mbps and 100 Mbps, Tcyc will scale to 400 ns ±40 ns and 40 ns ±4 ns respectively.
3. Duty cycle may be stretched or shrunk during speed changes or while transitioning to a received packet's clock domain as long as minimum duty cycle is not violated and stretching occurs for no more than three **Tcyc** of the lower of the transition speeds.

3.7.14 Transport stream interfaces

Table 3-42 Summary of TSIF support

Applicable standard	Feature exceptions	IPQ8064 variations
ITU-T H.222.0 Transport Stream (HTS); also known as ISO/IEC 13818-1	None	None

Table 3-43 Transport stream timing characteristics (96 MHz max)

Parameter	Comments	Min	Typ	Max	Unit
TSIF (clock frequency)		–	–	96	MHz
t(hddata)	Data hold	0.1	–	–	ns
t(hden)	Data enable hold	0.1	–	–	ns
t(hdsync)	Data sync hold	0.1	–	–	ns
t(sudata)	Data setup	4	–	–	ns
t(suenable)	Data enable setup	4	–	–	ns
t(susync)	Data sync setup	4	–	–	ns

3.8 Internal functions

Some internal functions require external interfaces to enable their operation. These include clock generation, modes and resets, and JTAG functions – as specified in the following subsections.

3.8.1 Clocks

Clocks that are specific to particular functions are addressed in the corresponding sections of this document. Others are specified here.

3.8.1.1 25 MHz core XO input

CLK_25M_OUT can be looped back on board to the CXO pin.

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3.8.1.2 Sleep clock

The 32 KHz digital sleep clock signal SLEEP_CLK_32K_OUT can be looped back on board to SLEEP_CLK.

3.8.1.3 PXO 25 MHz crystal

Connect a 25 MHz crystal across pins QSGMII_XTLI and QSGMII_XTLO. Parameters for the crystal are shown in Table 3-44.

Table 3-44 25 MHz crystal oscillator parameters

Symbol	Parameter	Min	Typ	Max	Unit
F_f	Frequency	–	25	–	MHz
F_S	Frequency stability over operating temperature @ 0-70 °C	-30	–	+30	ppm
F_t	Frequency tolerance @ 25 °C	-30	–	+30	ppm
F_O	Frequency tolerance	-50	–	+50	ppm
V_O	I/O voltage level	–	1.1	–	V
JitterRMS	Period broadband RMS jitter	–	–	1.5	ps
JitterPK-PK	Period broadband peak-to-peak jitter	–	–	50	ps
ESR	Series resistance	–	30	50	Ω
C_L	Load capacitance	–	15	–	pF
C_S	Shunt capacitance	–	7	–	pF

3.8.2 Modes and resets

The mode and reset functions are basic digital I/Os that meet the performance specifications presented in Section 3.4.

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3.8.3 JTAG

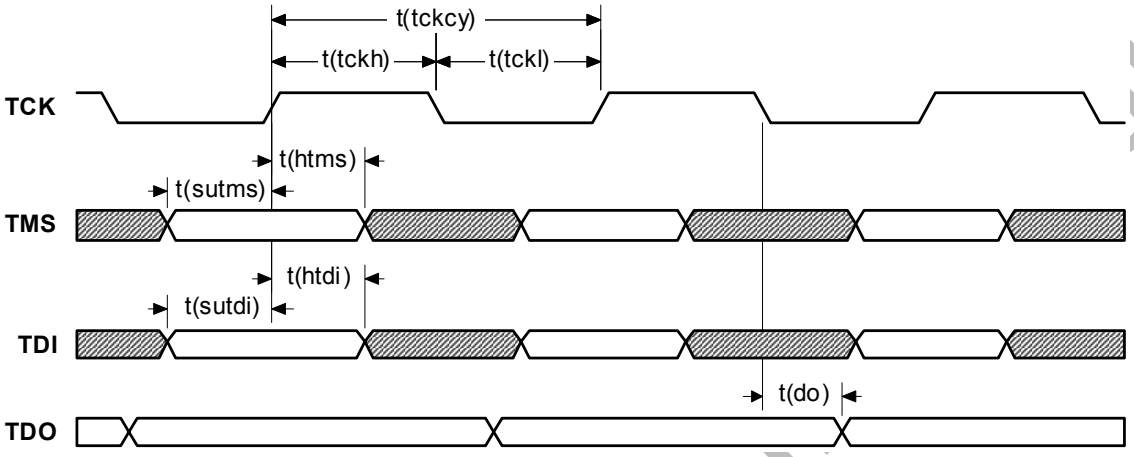


Figure 3-20 JTAG interface timing diagram

Table 3-45 JTAG interface timing characteristics

Parameter		Comments	Min	Typ	Max	Unit
t(tckcy)	TCK period		50	–	–	ns
t(tckh)	TCK pulse width high		20	–	–	ns
t(tckl)	TCK pulse width low		20	–	–	ns
t(sutms)	TMS input setup time		5	–	–	ns
t(htms)	TMS input hold time		20	–	–	ns
t(sutdi)	TDI input setup time		5	–	–	ns
t(htdi)	TDI input hold time		20	–	–	ns
t(do)	TDO data-output delay		–	–	15	ns

3.8.4 Serial Wire Debug (SWD)

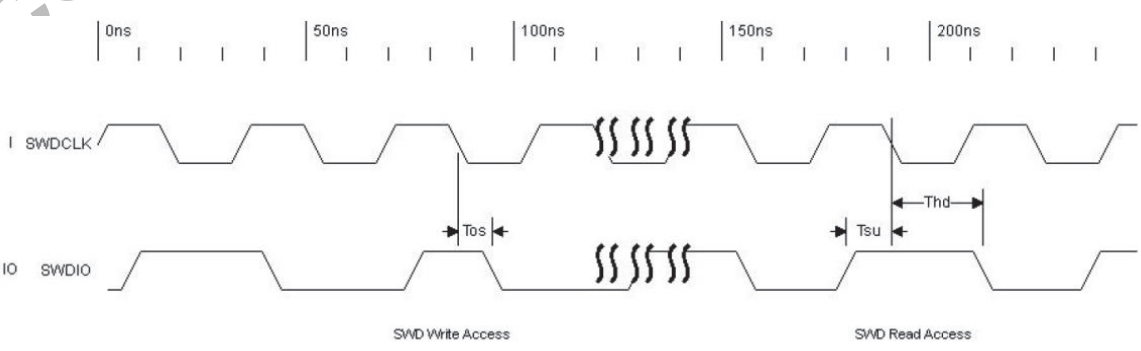


Figure 3-21 SWD write and read AC timing diagram

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Table 3-46 AC timing parameters

Parameter		Min	Max	Unit
T_{os}	SWDIO output skew to falling edge of SWDCLK	0	17.5	ns
T_{su}	Input setup time between SWDIO and rising edge of SWDCLK	4	--	ns
T_{hd}	Input hold time between SWDIO and rising edge of SWDCLK	1	--	ns

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4 Mechanical Information

The IPQ8064 uses flip-chip package technology as shown in Figure 4-1.

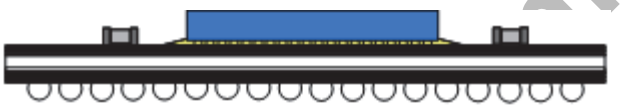


Figure 4-1 Flip-chip package cut-away diagram

4.1 Device physical dimensions

Figure 4-2, Figure 4-3, and Table 4-1 show the package drawings and dimensions for the 19×19 mm package.

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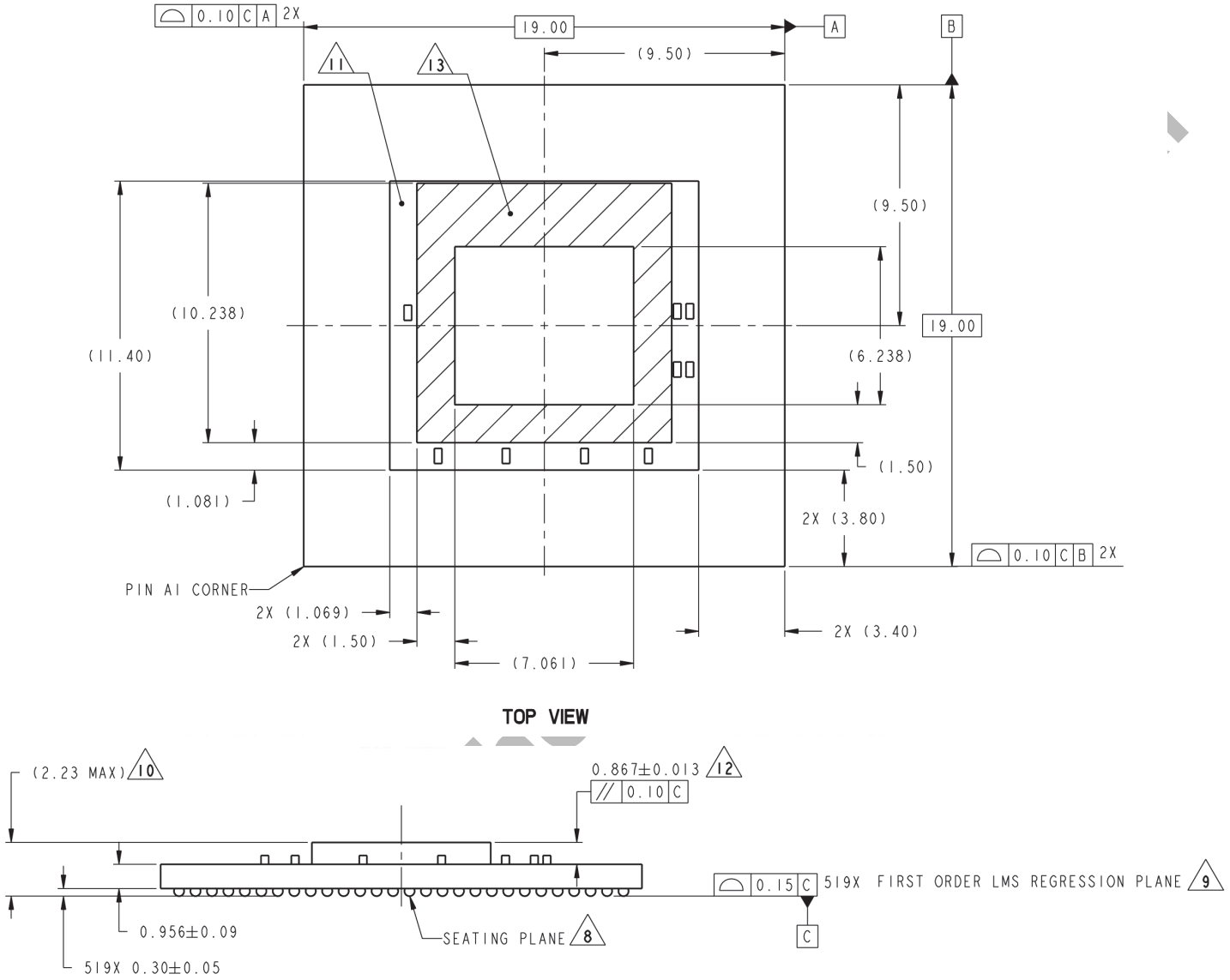


Figure 4-2 19x19 mm package details (top and side views)

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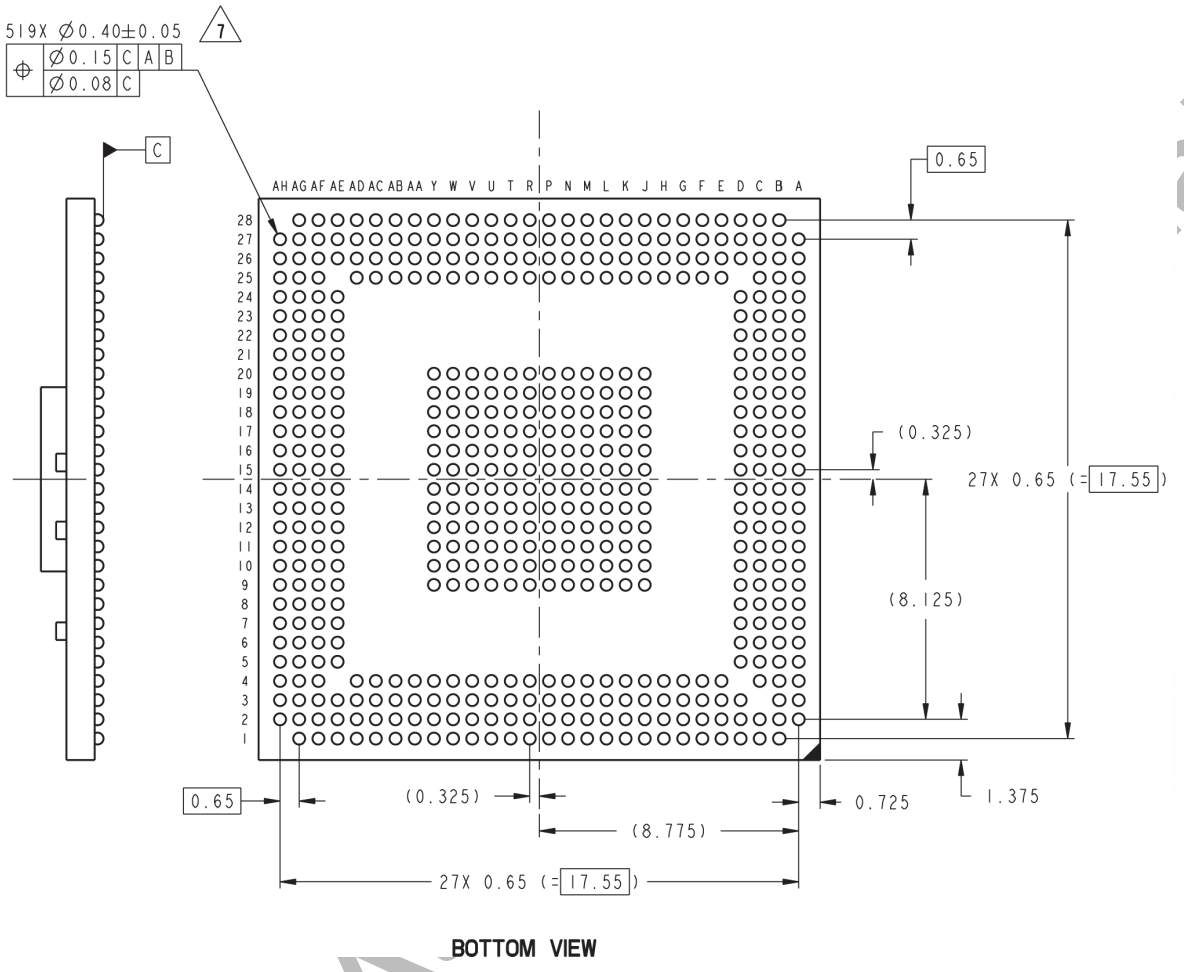


Figure 4-3 19x19 mm package details (bottom view)

Table 4-1 19x19 mm package notes

Note #	Note
1	Interpret drawing per ASME Y14.100.
2	All dimensions shown on this drowing are in millimeters (mm).
3	Interpret dimensions and tolerances per ASME Y14.5M-2009
4	Workmanship shall be in accordance with *Qualcomm Package Assembly Workmanship Standard* 80-V0691-2.
5	Qualcomm-supplied electronic database(s) are for reference only. dimensional information on current revision of released drowing takes precedence of electronic database(s).
6	Change approval. All changes shall be in accordance with 80-V3652-1 *General Supplier Quality Requirements*.
7	Dimension measured at the maximum solder ball diameter, parallel to the primary datum -C-.
8	The seating plane is defined by three non-colinear balls that support the free-standing package when it is placed on a flat surface. The vectors formed by the three balls establishing the seating plane shall include the center of gravity.

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Table 4-1 19x19 mm package notes (cont.)

Note #	Note
9	Primary datum -C- is determined by the first-order LMS regression plane through the spherical crowns of all solder balls on this side of the package.
10	Maximum package height determined by RSS tolerance method.
11	Allowable component area. Maximum component height shall not exceed 0.40 mm.
12	Dimension includes (0.08 mm) bump standoff with underfill.
13	Underfill area. Package contact prohibited.

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4.2 Part marking

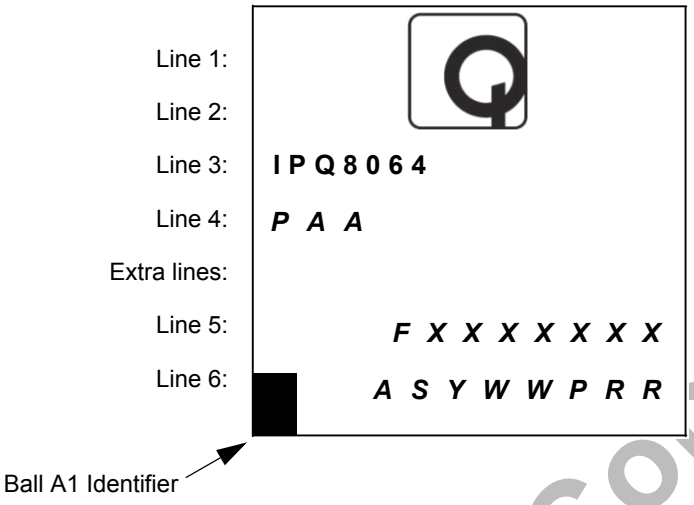


Figure 4-4 IPQ8064 marking (top view, not to scale)

Table 4-2 IPQ8064 marking line definitions

Line	Marking	Description
1 and 2	QUALCOMM	Qualcomm name or logo
3	IPQ8064	Qualcomm product name
4	*PAA*	P = product configuration code AA = product feature code
5	*FXXXXXXX*	F = supply source code XXXXXXX = traceability number
6	*ASYWWPRR*	A = assembly site code S = assembly sequence number Y = single, last digit of year WW = work week (based on calendar year) P = product configuration code RR = product revision
Additional lines may appear on the part marking for some samples; this is manufacturing information that is only relevant to Qualcomm and Qualcomm suppliers.		

4.3 Device ordering information

The order number is IPQ-8064-0-519FCBGA-MT-00-0.

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Order numbers have the form shown in Figure 4-5.

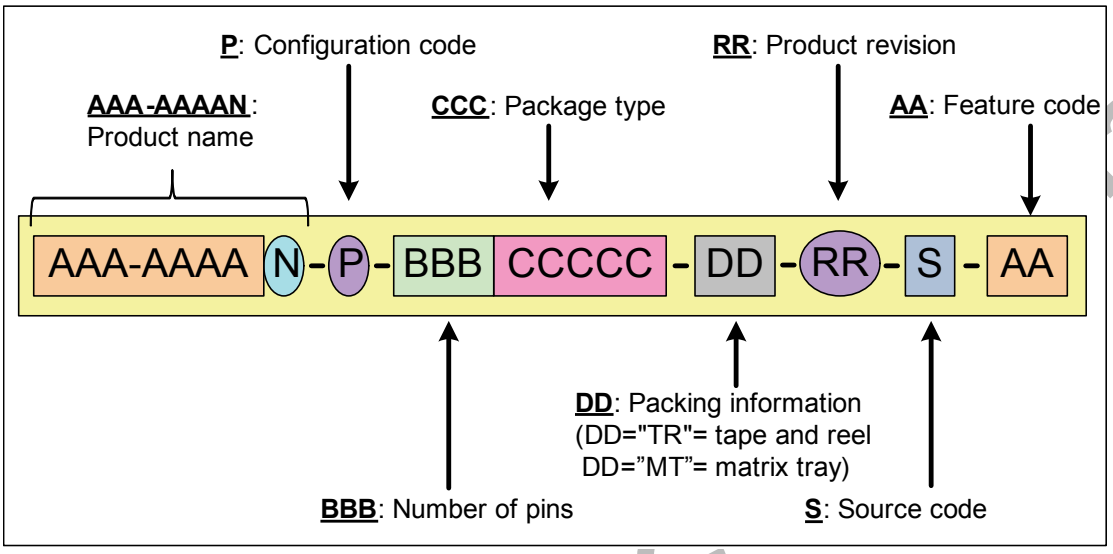


Figure 4-5 Device identification code

4.4 Device moisture-sensitivity level

Plastic-encapsulated surface mount packages are susceptible to damage induced by absorbed moisture and high temperature. Qualcomm follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. The IPQ8064 is classified as MSL3; the qualification temperature was 255ºC.

4.5 Thermal characteristics

Table 4-3 Thermal Resistance[1]

Parameter		Comment	Typical	Unit
θ_{JC}	Junction-to-Case	■ Ring-style cold-plate; insulated around package except for top ■ Assumes top-side cooling (i.e. heat sink)	0.21	°C/W
θ_{JB}	Junction-to-Board	■ Jedec JESD51-8 ■ Ring-style cold-plate ■ Maximum top case temperature used in calculation	9.10	°C/W
θ_{JA}	Junction-to-Ambient	■ Jedec JESD51-2A -Ambient Air chosen to be 25°C per standard ■ 1s2p test board ■ No thermal vias	18.39	°C/W

1. Contact your Qualcomm representative for power dissipation for each use case.

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5 Carrier, Storage, and Handling

5.1 Carrier

5.1.1 Tape and reel information

Carrier tape system conforms to EIA-481 standards.

A simplified sketch of the IPQ8064 tape carrier is shown in Figure 5-1, including the part orientation. Tape and reel details for the IPQ8064 are as follows:

- Reel diameter: 330 mm
- Hub size: 102 mm
- Tape width: 32mm
- Tape pocket pitch: 24mm
- Feed: Dual
- Orientation: Left (Q1)
- Units per reel: 1000

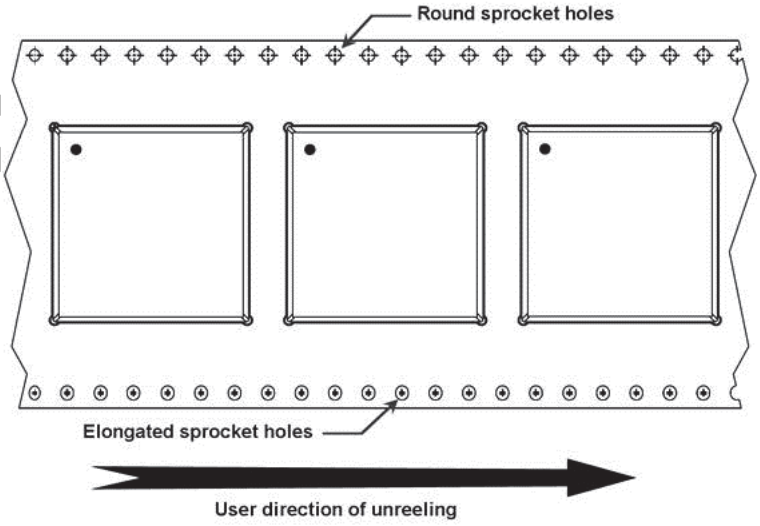


Figure 5-1 Carrier tape drawing with part orientation

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5.2 Storage

5.2.1 Bagged storage conditions

IPQ8064 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, anti-static bags. Refer to the *ASIC Packing Methods and Materials Specification* (80-VK055-1) for the expected shelf life.

5.2.2 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in Section 4.4.

5.3 Handling

Tape handling is described in Section 5.1.1. Other (IC-specific) handling guidelines are presented below.

5.3.1 Baking

It is **not necessary** to bake the IPQ8064 if the conditions specified in Section 5.2.1 and Section 5.2.2 have **not been exceeded**.

It is **necessary** to bake the IPQ8064 if any condition specified in Section 5.2.1 or Section 5.2.2 has **been exceeded**. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see *ASIC Packing Methods and Materials Specification* (80-VK055-1) for details.

CAUTION If baking is required, the devices must be transferred into trays that can be baked to at least 125°C. Devices should not be baked in tape and reel carriers at any temperature.

5.3.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

Products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, *Protection of Electrical and Electronic Parts, Assemblies, and Equipment*.

Refer to Section 7.1 for the ESD ratings.

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5.4 Barcode label and packing for shipment

Refer to the *ASIC Packing Methods and Materials Specification* (80-VK055-1) for all packing-related information, including barcode-label details.

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6 PCB Mounting Guidelines

6.1 RoHS compliance

The IPQ8064 complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC305 composition. A Product Material Declaration (PMD), which provides RoHS and other product environmental governance information, will be published when the data are available.

6.2 SMT parameters

This section describes board-level characterization process parameters. It is included to assist customers with their SMT process development; it is not intended to be a specification for their SMT processes.

6.2.1 Land pad and stencil design

The land pattern and stencil recommendations presented in this section are based on Qualcomm internal characterizations for lead-free solder pastes on an eight-layer PCB built primarily to the specifications described in JEDEC JESD22-B111.

Qualcomm recommends characterizing the land patterns according to each customer's processes, materials, equipment, stencil design, and reflow profile prior to PCB production. Optimizing the solder stencil pattern design and print process is critical to ensure print uniformity, decrease voiding, and increase board-level reliability.

General land pattern guidelines:

- Non-solder-mask-defined (NSMD) pads provide the best reliability.
- Keep the solderable area consistent for each pad, especially when mixing via-in-pad and non-via-in-pad in the same array.
- Avoid large solder mask openings over ground planes.
- Traces for external routing are recommended to be less than or equal to half the pad diameter to ensure consistent solder joint shapes.

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One key parameter that should be evaluated is the ratio of aperture area to sidewall area – the area ratio (AR).

For more information, refer to *PCB Land and Stencil Design Guide* (LS90-NG134-1).

6.2.2 Reflow profile

Reflow profile conditions typically used by Qualcomm for lead-free systems are listed in Table 6-1 and are shown in Figure 6-1.

Table 6-1 Typical SMT reflow profile conditions (for reference only)

Profile stage	Description	Temp range	Condition
Preheat	Initial ramp	< 150°C	3°C/sec max
Soak	Flux activation	150 to 190°C	60 to 75 sec
Ramp	Transition to liquidus (solder-paste melting point)	190 to 220°C	< 30 sec
Reflow	Time above liquidus	220 to 245°C [1]	50 to 70 sec
Cool down	Cool rate – ramp to ambient	< 220°C	6°C/sec max

1. During the reflow process, the recommended peak temperature is 245°C (minimum). This temperature should not be confused with the peak temperature reached during MSL testing, as described in Section 6.2.3.

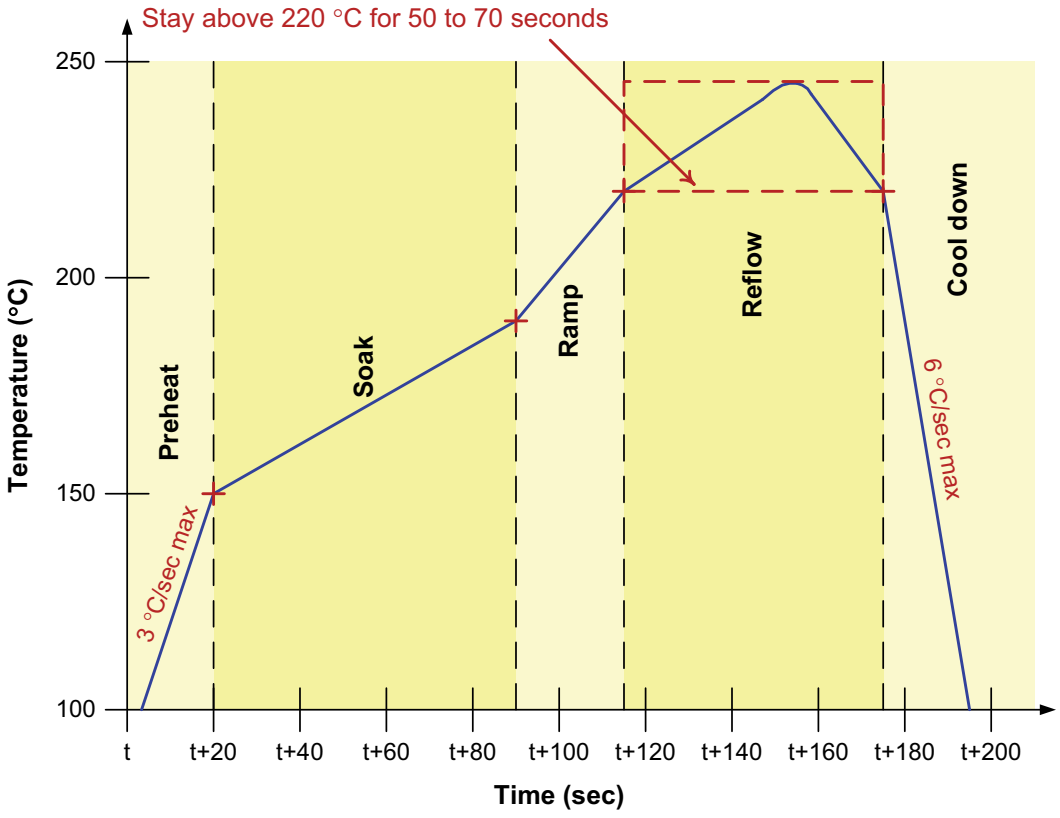


Figure 6-1 Typical SMT reflow profile

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6.2.3 SMT peak package-body temperature

This document states a peak package-body temperature in three other places within this document; without explanation, they may appear to conflict. The three places are listed below, along with an explanation of the stated value and its meaning within that section's context.

1. Section 4.4 – *Device moisture-sensitivity level*

   IPQ8064 devices are classified as MSL3@255°C. The temperature (255°C) included in this designation is the lower limit of the range stated for moisture resistance testing during the device qualification process, as explained in #2 below.

2. Section 7.1 – *Reliability qualification summary*

   One of the tests conducted for device qualification is the moisture resistance test. Qualcomm follows J-STD-020-C, and hits a peak reflow temperature that falls within the range of 260°C +0/-5°C (255°C to 260°C).

3. Section 6.2.2 – *Reflow profile*

   During a production board's reflow process, the temperature seen by the package must be controlled. Obviously the temperature must be high enough to melt the solder and provide reliable connections, but it must not go so high that the device might be damaged. The recommended peak temperature during production assembly is 245°C. This is comfortably above the solder melting point (220°C), yet well below the proven temperature reached during qualification (255°C or more).

6.2.4 SMT process verification

Qualcomm recommends verification of the SMT process prior to high-volume board assembly, including:

- In-line solder-paste deposition monitoring
- Reflow-profile measurement and verification
- Visual and x-ray inspection after soldering to confirm adequate alignment, solder voids, solder-ball shape, and solder bridging
- Cross-section inspection of solder joints for wetting, solder-ball shape, and voiding

6.3 Board-level reliability

Qualcomm conducts characterization tests to assess the device's board-level reliability, including the following physical tests on evaluation boards:

- Drop shock (JESD22-B111)
- Temperature cycling (JESD22-A104)
- Cyclic bend testing – optional (JESD22-B113)

For board-level reliability data, refer to *Board-Level Reliability 519 FCBGA* (BR80-Y0791-1).

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6.4 High-temperature warpage

Qualcomm measures high-temperature warpage using a shadow moire system. For detailed data, refer to *High-Temperature Warpage 519 FCBGA* (WR80-Y0791-1).

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7 Part Reliability

7.1 Reliability qualification summary

IPQ8064 reliability evaluation report.

Table 7-1 Silicon Reliability Results

Tests, standards, and conditions	Sample Size	Result
Mean time to failure (MTTF) t = 1000/λ in million hours (Total samples from three different wafer lots)	235	25 FIT; 40.00 million hours
ESD - Human-body model (HBM) rating ANSI/ESDA/JEDEC JS-001-2012 Target: 2000 V. (Total samples from one wafer lot)	3	PASS
ESD - Charge-device model (CDM) rating: JESD22-C101-E Target: 300 V per customer requirement (Total samples from one wafer lot)	3	PASS
Latch-up (I-test): JESD 78D Trigger current: ±100 mA; temperature: 85°C	3	PASS
Latch-up (Vsupply overvoltage): JESD 78D Trigger voltage: Each VDD pin, stress at 1.5 × Vdd max per device specification; temperature: 85°C	3	PASS

Table 7-2 Package Reliability Results

Tests, standards, and conditions	Sample size	Result
Preconditioning: J-STD-020, JESD22-A113F MSL 3, reflow temperature: 260°C+0/-5 Total samples from three different assembly lots	480	PASS
Temperature cycle: JESD22-A104D -55°C to +125°C; 1000 cycles 2 cycles per hour	240	PASS
Unbiased Highly Accelerated Stress Test: JESD22-A118A 130°C / 85% RH and 96 hrs duration	240	PASS
High-Temperature Storage Life: JESD22-A103D Temperature 150°C, 1000 hours duration	150	PASS

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7.2 Qualification sample description

Table 7-3 IPQ8064 characteristics

Device name	IPQ8064
Package type	FC-BGA
Package body size:	19 mm × 19 mm
Ball count	519
Ball composition:	SAC305
Process:	28 nm
Ball pitch	0.65 mm

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